

Group Coursework and Project Cover Sheet

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Group Design Project

AIR POLLUTION AND ECONOMIC ACTIVITY

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1. Background

1.1 Introduction

Measurement of economic activity in a region is important to understand the economic well-being of its people and help plan future policies. The common method to measure and express economic activity is using Gross Domestic Product (GDP). In many low- and middle-income countries, particularly at the subnational level, GDP statistics are not produced at the speed, spatial resolution, or level of trust required for real-time policy and planning and may remain limited or unavailable. Furthermore, in conflict-affected or resource-constrained regions such as Addis Ababa and Baghdad, national accounts are often delayed, incomplete, or subject to manipulation, making it difficult to track economic performance or design responsive policy. In response, researchers increasingly rely on high-frequency proxy indicators derived from satellite and sensor data that reflect real-time patterns of economic activity.

This project, Air Pollution and Economic Activity, explores whether tropospheric nitrogen dioxide (NO₂), a short-lived combustion byproduct released from vehicles, power plants, and industrial processes, can serve as one such proxy. NO₂ is measurable from space using instruments like Sentinel-5P, Plume Labs, and OpenAQ, and its strong link to fossil fuel combustion makes it a promising candidate for tracking economic intensity at high temporal resolution. We hypothesise that NO₂, when interpreted alongside established proxies such as nighttime lights and surface groups, can offer a more comprehensive, spatially detailed, and timely view of subnational economic trends in regions where official GDP data are weak.

1.2 Research Objectives

The research aims to address the following objectives by researching the existing literature and available data.

1. To identify and analyse the key factors influencing nitrogen dioxide production and measurement.
2. To evaluate the extent to which satellite-derived nitrogen dioxide levels can be spatially and temporally attributed to the main NO₂ contributors of industrial pollution and mobility.
3. To review and apply relevant methods and datasets for validating the relationship between NO₂ levels and economic activity, particularly in Baghdad and Addis Ababa.
4. To critically assess the limitations and underlying assumptions of using NO₂ and nighttime lights as proxies for studying trends in economic activity.

2. Literature Review

2.1 Case Studies

Existing projects from the World Bank Group highlight case studies from Baghdad, Iraq and Addis Ababa, Ethiopia. This project will focus on these regions and attempt to supplement these ongoing studies.

2.1.1 Addis Ababa

Often considered the political capital of Africa, Addis Ababa is the capital of Ethiopia. As shown in Figure 2.1, it is the geographical centre of Ethiopia as well as the central political, economic, and cultural hub. Tabor (2019) noted that, as a major contributor to Ethiopia's GDP, the city's development has been given special focus by past governments as compared to other regions in the country. Furthermore, Addis Ababa has continuously been undergoing horizontal expansion over the past decades, leading to loss of vegetation and agricultural land. Migration from other regions, especially rural areas, has further stressed the local resources, particularly in the inner zone.

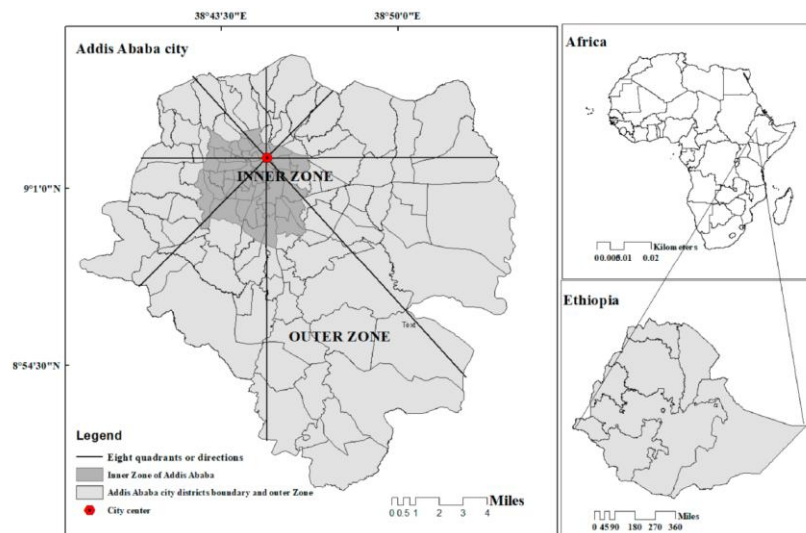


Fig. 2.1 Addis Ababa City

Credit: Woldeamayrat and Genovese (2021)

As noted by the World Bank in the 2019 document on Integrated Strategic Development for Addis Ababa, low density urban expansion has led to increased private vehicle commuting in the city. Additionally, lax vehicle emission control and an aging fleet along with indoor cooking and heating, industrial pollution, and open trash burning have all contributed to increasing air pollution in this city.

The same document further notes that the unmanaged urban sprawl has led to increased traffic congestion in Addis Ababa. The spatial mismatch between jobs and housing, inadequate traffic enforcement, parking supply and traffic management system has exacerbated the traffic congestion problem. Limited public transportation, concentration of urban roads in the inner zone, and lack of pedestrian roads have made transportation inaccessible to many people. A 2018 World Bank document on economic development in Addis Ababa by Hobson showed that

the major industries in Ethiopia experiencing a growth in employment are agro-processing, publishing, textiles and clothing, leather and footwear, furniture, metals, and plastics. Addis Ababa has the largest production share in most of these sectors and accounts for nearly 29% of GDP in Ethiopia (Weldeghebrael, 2021). Another growing industry in the city is the services sector, including retail, transport, and tourism.

In Addis Ababa, the Ethiopia Country Economists' Office aims to use nighttime lights and air quality data to detect signs of economic change following recent policy reforms.

2.1.2 Baghdad

The Baghdad Governorate, or Baghdad Province, is home to Baghdad City, the capital of Iraq. It is the cultural, manufacturing, and transportation hub of the country. Al-Hameedi et al. (2021) showed that over the past four decades, urban land has been growing in Baghdad while agricultural land and natural vegetation have decreased considerably. For decades, the country of Iraq has been ravaged by war, conflicts, and terrorism, the latest being the Iraq War from 2003 to 2011 and the ISIL insurgency between 2013 and 2017. This has led to large numbers of rural populations fleeing to other countries, and some to urban areas of Baghdad City (Al-Hameedi et al., 2021). Thus, there is an imbalance in the population distribution in Baghdad.

As noted by Chaichan et al. (2016), the number of gasoline locomotives has increased significantly in Iraq since 2003 and more than a third of these are concentrated in Baghdad. They found that diesel-fueled trucks, buses, heavy-duty construction equipment, and diesel electricity generators are all contributing to the deterioration of air quality in Baghdad. Furthermore, the popular use of heavy fuel oils in power generation and large amounts of sulphur in fuels is also contributing to air pollution. High levels of NO₂ were found near highways.

The prolonged conflicts in Iraq have led to severe infrastructural damage, social and political instability, and impoverishment over the years. However, owing to its vast oil reserves, the country has been able to become an upper-middle-income country. The Iraqi Economists Network (2020) noted that despite being a highly oil-dependent country, notable industries in Iraq other than oil and gas include agriculture, food and beverage (soft drinks, dates), metal processing, and furniture. Several small industrial establishments are active in Iraq; they are involved in bakery products, construction metal works, furniture, and clothing. Nearly a fifth of the industrial establishments in the country are in Baghdad.

An article by Saadawi et al. (2025) discussed the reasons for and effects of traffic congestion in Baghdad. Notably, the severe congestion in the city is some of the worst in the Middle East and has been caused by deteriorating roads, numerous security checkpoints, increasing car numbers, inadequate public transportation, and delays and corruption in infrastructure projects. The congestion has led to longer travel and wait times causing excess fuel consumption as well as reduced footfall at local businesses. It was estimated that traffic congestion in Baghdad costs the Iraqi economy about \$2 billion annually.

In Baghdad, transportation planners are looking to use Waze traffic jams data to understand mobility's broader economic and social impacts.

2.2 Nitrogen Dioxide

Nitrogen dioxide is a reddish brown acidic oxidizing gas with a pungent, irritating odour, produced through combustion and nitric oxide (NO) oxidation. Nitric oxide itself is produced by both natural and anthropogenic sources. The process of combustion, i.e. burning of fossil fuels in vehicles, industries, and power plants, produces both NO₂ and NO, which are together referred to as NO_x. Nitrogen dioxide is a major component of vehicle exhaust and is often used as a proxy to measure traffic-related air pollution. Other sources of NO₂ include diesel-powered heavy equipment, kerosene or gas heaters, industrial boilers, industrial emissions, lightning, soil, and fires.

The effects of nitrogen dioxide on human health include reduced lung function, inflammation of the airways, and increased asthma attacks, with pregnant individuals and minors at increased risk. NO₂ also produces another pollutant, ozone, and contributes to acid rain, both of which are detrimental to the environment; thus, it is a regulated gas that is constantly being monitored. Due to its short lifetime (less than a day), most of the NO₂ in the atmosphere is located near the ground, in the troposphere, where living beings live and breathe.

The methods of measuring NO₂ can be broadly classified as ground-based methods and satellite remote sensing. Ground-based measurement methods are used for ambient monitoring and include chemiluminescence analysers, passive diffusion tubes, electrochemical sensors, thick film sensors, and differential optical absorption spectroscopy (DOAS). DOAS may also be employed from aircraft, balloons, and even satellites. These methods have high accuracy but large gaps in coverage due to limited availability and adoption across different countries. Chemiluminescence analysers are the standard for the US EPA's Air Quality System (AQS) and are mandated by Directive 2008/50/EC for regulatory monitoring in the EU. A summary of these methods based on the publications of DEFRA is given in Table 2.1.

In the context of this study, a greater focus is placed on remote sensing using satellites for NO₂ measurement. This is because satellite data is available globally and is often open access. It is especially beneficial for data poor countries where ground data may be unavailable due to inaccessibility, warfare, technological limitations, political instability, etc. Ezran et al. (2023) point out that while ground stations for measuring air pollution may be unevenly and sparsely located, satellites provide data from multiple instruments at the same time each day.

A popular satellite used for nitrogen dioxide measurement is Sentinel-5 Precursor, which is operated by the European Space Agency (ESA) and the Netherlands Space Office (NSO). It uses the Tropospheric Monitoring Instrument (TROPOMI) to monitor atmospheric NO₂ levels globally. TROPOMI is a nadir-viewing, imaging spectrometer that measures the intensity of sunlight scattered back from the Earth's atmosphere.

According to the Algorithm Theoretical Basis Document (ATBD) for TROPOMI by van Geffen et al. (2019), it calculates the total NO₂ in a vertical column of the atmosphere and uses advanced algorithms to distinguish stratospheric and tropospheric NO₂ with a native spatial resolution of 3.5 x 5.5 km² and daily coverage. This spatial resolution is reprocessed to obtain Level-3 products having a resolution of 1113.2 x 1113.2 m². Its high spatial resolution and signal-to-noise ratio make TROPOMI suitable for measuring stratospheric NO₂. TROPOMI uses

DOAS retrieval to determine the NO₂ slant column density (SCD). SCD represents the total amount of NO₂ along the line of sight. DOAS retrieval is the first step in NO₂ processing. The tropospheric SCD is obtained by subtracting the stratospheric SCD from the total slant column. Then, the two slant sub-columns are converted to vertical NO₂ columns. For stratospheric NO₂, the bias is found to be less than 10%, while for tropospheric NO₂ it is about 25-50%. These uncertainties are due to both retrieval errors and instrument errors. Retrieval errors, especially in polluted areas, may be due to the presence of clouds, aerosols, and the surface albedo. Similarly, in rural areas, the NO₂ concentrations are often low, but errors in measurement may be due to random noise due to the instrument's Signal-to-Noise ratio (SNR).

Table 2.1 Ground-based NO₂ Measurement Methods

Method	Description	Limitation
Chemiluminescence	The analyser makes two independent measurements of NO and NO _x (NO+NO ₂), either sequentially or using dual chambers. It has a low detection limit of $\sim 1 \mu\text{g m}^{-3}$.	High operating and capital costs.
Diffusion Tubes	It uses colorimetry and a tube with stainless steel grids coated with triethanolamine that adsorbs NO ₂ to produce a nitrite salt. It has low capital and operating costs.	Accuracy depends on the method of tube preparation and the analysing laboratory.
Electrochemical Sensors	It uses portable samplers used to perform the electrochemical reduction of NO ₂ entering from the air. It is widely used for assessing occupational exposure.	Higher detection limit of some samplers ($\sim 200 \mu\text{g m}^{-3}$).
Thick Film Sensors	It uses portable samplers with nano-structured semi-conducting metal oxides. It has the potential to identify pollution hotspots.	Still under research for commercial purposes.
DOAS	The light received is analysed in an opto-analyser system using the Beer-Lambert Law. It has a low detection limit of $\sim 1 \mu\text{g m}^{-3}$. It can also be employed on satellites.	Weather conditions can affect instrument performance.

As highlighted by Follette-Cook et al. (2020), the amount of NO₂ in the troposphere is not only determined by emissions from anthropogenic activities, such as combustion of fossil fuels in cars, power plants, and manufacturing, but also by natural emissions from lightning, fires, along with atmospheric chemistry, and weather. It is also produced during welding and using explosives, refining petrol, commercial manufacturing, and food manufacturing. Weather effects such as winds can disperse NO₂, temperature and sunlight can affect the speed of NO₂ chemistry in the atmosphere, while rain can reduce the persistence of NO₂ in the air by washing it away. A major limitation, as highlighted by Geddes et al. (2012), of using remote sensing is the occurrence of data gaps or inaccuracies due to cloud cover. The magnitude of bias has been

found to be location-specific, varying across urban, suburban, and rural sites. For the regions of Addis Ababa and Baghdad, this gap has been addressed by iterative interpolation, as shown in Chapter 3. Removing the data affected by cloud cover can introduce selection bias. Furthermore, averaging NO₂ level values as daily values excludes diurnal variation. When using NO₂ levels as a measure of economic activity, this bias and exclusion of variation can interfere with the estimation of true economic activity levels. There are also limitations based on the source chosen for data collection, for instance, TROPOMI is more responsive to NO₂ concentrations in the upper troposphere and stratosphere than near the surface, as evidenced by Verhoelst et al.

2.3 Economic Activity

Gross Domestic Product (GDP) is widely used as an indicator of a country's welfare. However, GDP measurement in developing countries is often inadequate due to their smaller formal sector, limited regional integration, price uniformity, and weaker national statistical systems (Henderson et al., 2012). Baghdad and Addis Ababa are both situated within developing countries, yet they differ significantly in terms of income levels and economic structure. Iraq is classified as an upper-middle-income country, largely owing to its substantial oil reserves, which contribute significantly to national revenue. However, as noted by the UNDP, Iraq is the most oil-dependent country globally and suffers from poverty and unemployment, specifically among youth and women. On the other hand, Ethiopia is a low-income country, where the majority of the population relies on agriculture and informal employment.

The existing literature was assessed for a variety of high-frequency indicators to fill subnational economic statistics gaps. We assessed a suite of proxies based on data availability, validation performance, and sectoral coverage. Where detailed spatial income data are unavailable, Henderson et al. (2012) illustrated that nighttime lights (NTL) data could provide valuable insights for analysing economic growth at both subnational and supranational scales. Similarly, Ghosh et al. (2010) found a direct positive relationship between nightlights and GDP in countries and subnational units, including low- and middle-income countries that possess very low-quality national accounts data. Areas with high nightlight intensity tend to coincide with higher population density, increased infrastructure investment, and higher energy consumption.

People's mobility is one of the proxies considered. Captured by Google Community Mobility Reports and Apple Maps routing, it reveals near-real-time shifts in retail footfall, workplace attendance, and transit usage. Putra & Arini (2020) demonstrate that once adjusted for regional industry composition, mobility fluctuations explain up to 84% of quarterly regional GDP variability across Indonesia's provinces, far outperforming simple regressions on mobility alone ($R^2 \approx 0.74$). Moreover, mobility is endogenously linked to NO₂ emissions. It is driven by vehicle use and biased toward digitally connected urban populations, limiting its independence as a benchmark for combustion-based proxies. Thus, it can still be applied to measure mobility in cities where there is data availability.

Waze traffic jams data, derived from GPS location data from Waze application users and crowd-sourced incident reports, estimates deviations from free-flow traffic speeds and categorises jams into six severity levels (Waze Partners Help, 2025). However, user coverage of Waze data is neither uniform nor verified due to the lack of information on how Waze usage varies by day,

location, or demographic group. This introduces potential bias, especially in areas with low app penetration or where usage habits are not representative of the broader population (Gu, 2022). Another existing bias for Waze data is its category of “unusual traffic”, and the classification of jam severity is not fully transparent, making it difficult to assess the reliability of these indicators as proxies for congestion or economic activity (Gu, 2022).

Another natural candidate is electricity consumption (kWh drawn from the grid), which similarly tracks industrial output and household activity. Bundervoet et al. (2015) found a solid elasticity (~ 0.37) between electricity use and GDP in sub-Saharan Africa, but one notably lower than the elasticity for nighttime lights (0.57), in part because off-grid generators and informal power sources go uncounted in official data. Moreover, electricity metrics miss non-electrified rural sectors and suffer from reporting lags and outages, making them less reliable in rapidly changing contexts. Energy consumption is not selected as a proxy for economic activity in this project due to the lack of direct energy price data for the regions under study, which may result in an inaccurate reflection of actual energy costs. In addition, the identified patterns of causality may not fully account for the structural or institutional factors that could explain cross-country variation, which requires further investigation (Akinlo, 2008).

High-resolution gridded population maps such as LandScan and WorldPop offer yet another proxy. According to Wang et al. (2022), by allocating a country’s aggregate GDP uniformly across population-weighted grid cells, these maps serve as a baseline for subnational disaggregation. When combined with NTL in the “LitPop” method, population data elevate subnational R^2 from ~ 0.75 (NTL alone) to over 0.90, vastly improving spatial GDP allocation across China, the U.S., and hundreds of global provinces. Nevertheless, population data inherently assume uniform per-capita productivity and fail to reflect short-run economic shocks such as sudden factory closures or mobility restrictions.

Land use plays an important role in rapid and often informal urban development. Ahn et al. (2023) attempted to model economic development in the data-poor country of North Korea by using a human-machine collaborative approach using surface cover satellite imagery. In the absence of ground truths, they used deep learning to cluster the images obtained from a grid of satellite imagery based on common land pattern features such as urban, rural, and uninhabited. The approach used public satellite imagery and minimal human input for subjective ranking to measure economic activity. Their results suggested that areas far away from major cities are associated with less development, which makes intuitive sense. They also found that Economic Development Zones (EDZ) designated for agriculture or tourism developed more as compared to non-EDZ regions. Chen et al. (2020) note that regional economic development is strongly related to land use and land cover change (LUCC) in developing countries such as China. Similarly, information obtained from remotely sensed imagery, such as Enhanced Vegetation Index (EVI) data, were used to analyse the spatiotemporal patterns and driving mechanisms of vegetation change (Wang et al., 2023). EVI offers a robust way to monitor agricultural activity and land-use change. EVI is derived from MODIS reflectance data and improves upon NDVI by reducing atmospheric distortion and providing more sensitivity in dense vegetation. This makes it particularly suitable for tracking crop cycles and vegetation health in agrarian zones such as the rural outskirts of Addis Ababa. As Ahn et al. (2023) and Wang et al. (2023) show, EVI can reveal economic shifts in areas that NTL fails to capture, especially where agriculture

drives livelihoods, but electrification is limited. These products directly measure crop cycles and land-use changes, providing annual or seasonal signals of primary-sector performance that lights cannot detect. In predominantly agrarian regions surrounding Addis Ababa's periphery, these surface-cover metrics ensure our proxy framework captures the whole economy, from farms to factories to offices.

Similarly, Lehnert et al. (2023) used daytime satellite imagery as a proxy for measuring economic activity. They used surface groups to define the land cover as built-up, grass, crops, forest, no vegetation, and water. Meanwhile, the function of economic and social infrastructure, such as shopping malls, logistics centres, offices, and public facilities, attracts regular traffic and economic demand. By analysing the relationship between infrastructure levels, travel distances (mobility activity), and human development metrics (such as HDI, SDGs, and the Social Progress Index), it is demonstrated that increased mobility infrastructure and activity are linked to economic growth and enhanced well-being (Virág et al., 2022). Thus, surface groups along with economic zones, such as the Special Economic Zones (SEZ) of Ethiopia, and places of interest such as roads, banks (Hasan et al. 2018), airports (e.g. Baghdad International Airport) can be used to measure economic activity.

Ahn et al. (2023) also point out that since cloud cover can interfere with satellite imagery, its presence may mislead the model. Furthermore, satellite imagery does not account for factors such as factory utilisation, traffic, and underground operations. The usage of satellite imagery also means that there may be a temporal mismatch with actual economic events as shocks to the economy may not necessarily be reflected by change in land use until after some time. Intra-urban economic variation may also not be captured as all the area will be categorised as built-up area. Furthermore, the variation in individual output, value added to the economy, and employment created by different firms, factories, and plants will not be covered by this approach for measuring economic development.

2.4 Use of NO₂ to Measure Economic Activity

This project aims to assess the reliability of using NO₂ as a proxy for measuring economic activity. Ezran et al. (2023) proposed using satellite readings of NO₂ to improve the measurement of global economic activity and found it particularly useful in countries facing data manipulation, conflict, or other factors leading to poor national accounts. They illustrated that as NO₂ stays in place and has a short atmospheric life; it is directly indicative of the combustion below it. This makes NO₂ a preferable signal of economic activity. Furthermore, they showed that NO₂ resolves significant GDP errors when dealing with hyperinflation, conflict, fluctuating oil prices, and financial crises. It is reliable when considering cyclical fluctuations, which may be missed out on by other satellite estimates such as nightlights. However, when it comes to industrial production, there may be heterogeneity in production technology across regions and across time. This could mean that NO₂ may be a better proxy for measuring economic activity in some cases than others.

In this study, an assumption is made in using NO₂ as a proxy for economic activity; the changes in NO₂ concentrations are primarily driven by shifts in transportation and industrial combustion. This assumes a direct and consistent link between human mobility and industrial production with emissions, which may not hold in all settings, particularly where non-economic events or

unmonitored private generators play a role. Vertical and horizontal transport due to weather can cause NO₂ to drift from the emission source, reducing spatial precision. Ji et al. (2018) mention that satellite data captures column densities and not ground-level exposure, which may misrepresent activity on the surface. Thus, NO₂ may be supplemented with other data, such as land use cover, to obtain a better understanding of economic activity, particularly in the case of developing or fragile countries.

2.5 Conclusion

While the proxies above each offer valuable insights, none single-handedly captures the economic activity due to both industrial production and mobility activity with both spatial and temporal fidelity.

Nightlights underrepresent non-electrified agricultural areas, and the data is susceptible to missing values due to cloud cover and stray light. Nitrogen dioxide is a fairly new but highly promising indicator of industrial production and congestion. However, given its limitations, we look at other supporting variables as indicators of economic activity such as surface-cover and places of interest, particularly those where significant commercial activities take place. After a review of the various possible indicators of industrial production and mobility, a list of variables of interest is obtained as shown in Table 2.2. Nightlights have been included in this table as the applicability of NTL data to the regions of interest will be tested for completeness.

Table 2.2 Variables of Interest

Type		Name	Description	Source
Air Pollution		NO ₂ Levels	Daily NO ₂ data in mole/m ² at mesh-level	TROPOMI (Sentinel-5P)
		Air Pollutants	Monthly NO ₂ , SO ₂ , CO, O ₃ data at city-level	Sentinel-5P
		Cloud Cover	Clear or cloudy attribute per cell	NASA MODIS Aqua
		Surface Temperature	Daily temperature value spatially averaged for each cell (°C) at 10:30 AM (LST morning) and 1:30 PM (LST day)	NASA MODIS Aqua + Terra
Economic Activity Indicators	For Industrial Production	Nightlights	Daily data in nW/cm ² /sr at 15 arc-sec resolution (approx. 500 m)	NASA Black Marble
		Exports	Ethiopian exports in 2023 and 2024 in Birr per month	Ethiopian Statistical Service
		Surface Groups	Land use data at upto 10 m resolution (industrial area, residential area, etc.)	Sentinel-2, ESA, and OpenStreetMap
	For Mobility	Points of Interest	Percentage of places producing considerable NO ₂ due to commercial activity/mobility (supermarket, hospital, car_dealership, car_rental, car_wash, market_place, mall, university)	OpenStreetMap
		Traffic Congestion Intensity (TCI)	Value based on number of traffic jams, vehicle speeds, etc.	Waze
		Road length	Total km of road length per cell in the mesh for 2023-24	OpenStreetMap
		Distance to road	Distance to primary road in metres	OpenStreetMap
	Supplementary Variables	Population	Total population per cell in the mesh for 2023-24	WorldPop
		Boundary	City boundary for Baghdad and Addis Ababa (latest)	HDX

3. Data Processing

3.1 Dataset Description

To build the NO₂ model and economic activity model, the following aspects are considered, including socio-economic data, mobility data, geographical data, climate data, emission data, territorial spatial data, and economic activity data. Descriptions of the datasets are as outlined in Table 2.2.

The following assumptions are used before data preprocessing:

1. The data does not include cloud-covered scenes, nor the scenes covered by snow/ice, and errors or problematic retrievals.
2. Data obtained from Sentinel-5P is considered reliable due to lack of ground truth data (eg. OpenAQ, PlumeLabs etc.) for benchmarking.

3.2 Data Cleaning and Aggregation

Each of the data sources is processed through the following steps:

Step 1: Prepare boundaries and address missing values

1. Boundary files: The geographical boundaries for Baghdad and Addis Ababa are prepared in ArcGIS by converting raster data to a polygon layer.
2. Missing values: The missing values are addressed for each data source using iterative filling, using mean of the neighbour raster as the replenish value.

Step 2: Clip the data source to the regions of interest

Data related to the regions of interest is filtered using the selected boundary polygon (Baghdad or Addis Ababa).

Step 3: Spatial aggregation to mesh

Aggregate Values: Zonal statistics are calculated within each mesh cell, and a mean or sum of the data is calculated based on data source characteristics (e.g. the sum of population is calculated considering data about population). The aggregation methods required to complete the analysis are outlined as follows in Table 3.1.

Step 4: Visualisation

Generate visual outputs using the aggregated mesh data: static maps for spatial distribution and time series plots for temporal dynamics.

As missing values are filled using iterative filling, mean values are used for aggregation for both nightlight intensity and NO₂. This is done to avoid proliferation of values that are not based on real-time measurements and thereby preserve the authenticity of the data. A summary of the aggregation methods chosen is given in Table 3.1.

Table 3.1 Aggregation Methods for Datasets

Dataset Name	Aggregation Method	
	At cell level	At sub-city or district level
Population	Sum	Sum
Traffic Congestion Data	Sum	Sum
Road Length	Sum	Sum
Distance to Primary Road	Minimum	/
Point of Interest, POI	Total Count	Total Count
NO _x Concentration	Mean (spatial)	Sum
Country Boundary	/	/
Air Pollutants	Sum	Sum
Nightlight Intensity	Mean (spatial)	Sum
Land Use	Total Area	Total Area
Cloud Cover	Mode	/
Surface Temperature	Mean (spatial)	Mean (spatial)

3.3 Data Availability and Reliability

Most of the data used in this project is publicly available, with a significant portion sourced from OpenStreetMap and other open platforms such as WorldPop and Sentinel-5P. While most of the data is free upon request and generally extensive, there are notable differences in time range and time resolution across sources. Some datasets provide yearly averages, while others offer more detailed time ranges, such as monthly averages or daily values, making integration and comparison across datasets more complex. Datasets come in various formats, such as GeoTiff, CSV, and shapefile, and are outputted as JPQ.

The selected data is deemed reliable, as it is sourced from various widely used open platforms. However, the reliability of export data from the Ethiopian Statistical Service depends on national accounts. The coverage of OpenStreetMap (OSM) is contingent upon local contributor activity.

3.3.1 Limitations and Biases

The Sentinel-5P TROPOMI Level-3 NO₂ product arrives fully geolocated and pre-calibrated, having undergone extensive on-orbit corrections at the Level-1b stage. These include:

1. Wavelength alignment via solar Fraunhofer line fitting to correct thermal drifts (residual uncertainty ≈ 0.01 nm, inducing slant-column errors of 5–10%).
2. Stray-light (Ring) subtraction using a lookup table and polynomial model, which can under- or over-correct scattered light by $\pm 5\%$ depending on scene brightness.
3. Radiometric flat-fielding from onboard lamp and solar calibrations, with residual detector ageing drift of 2–5% over multiple months.
4. DOAS spectral fitting choices (fit window, polynomial order, absorption cross-sections) that impart 3–8% variability in retrieved slant columns.
5. Air-mass factor (AMF) assumptions, converting slant to vertical columns based on modelled NO₂ profiles, surface albedo, clouds, and aerosols; each contributing 5–20% uncertainty in heavily polluted boundary layers.
6. Row-anomaly correction (“de-striping”) that can inadvertently remove genuine longitudinal gradients if anomalies coincide with real plumes.
7. Quality filtering using `qa_value > 0.75` to exclude cloudy or contaminated pixels, which can bias results toward clearer conditions.

Although these calibrations occur upstream, their residual biases persist in the Level-3 product. In Addis Ababa, high altitude and frequent cloud cover can skew AMF assumptions, leading to up to 15% underestimation of column burden during moist seasons. In Baghdad, desert dust and high aerosol loading often cause stray light and cloud flags to misclassify clear scenes as cloudy, resulting in seasonal gaps and potential low bias in NO₂ estimates.

The disaggregation of exports data is based on a list of 100 industries in Addis Ababa and their sales revenue. It is assumed that this disaggregation is reflective of each sub-city’s contribution to Addis Ababa’s industrial production. As Weldeghebrael (2021) stated that Addis Ababa contributes to 29% of Ethiopia’s GDP, it is assumed that the city contributes similarly to national exports.

4. Exploratory Data Analysis

4.1 Addis Ababa

4.1.1 NO₂

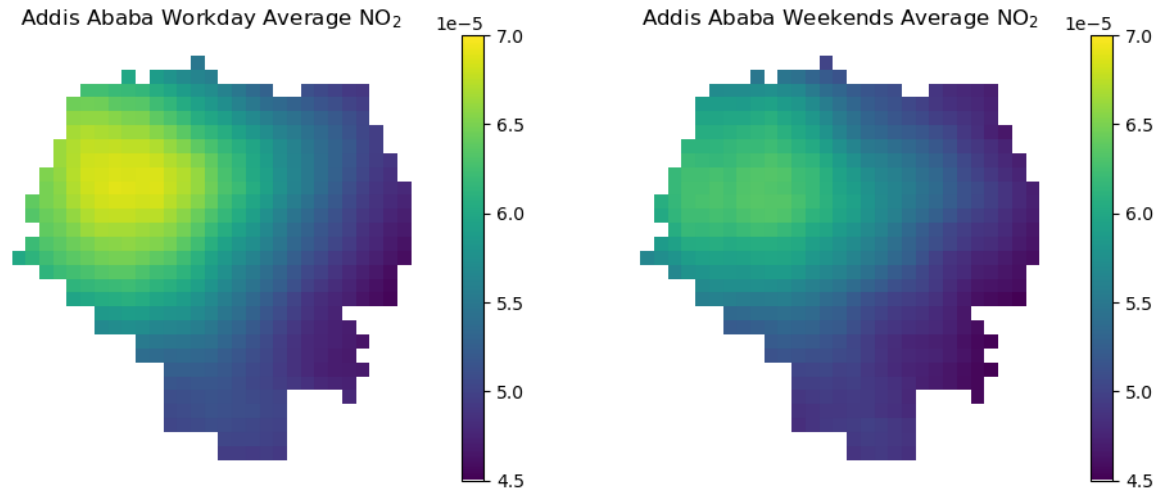


Fig. 4.1 Addis Ababa Average NO₂ Levels (in mole/m²)

For both weekdays and weekends, there is a prominent concentration of NO₂ in the northwestern region. This region, highlighted in pink in Figure 4.3, is the heart of Addis Ababa. It is where most of the roads converge. Most of the Addis Ababa Light Rail System is located here as evidenced by Figure 4.2. The Addis Ababa Bole International (ADD) Airport is located towards the southeast of this region as shown in Figure 4.3. The figure also shows the major highway corridors of A1 and A2. The A1 highway connects Addis Ababa to Dire Dawa and Djibouti, and it is a part of the Trans African Highway network. These mobility systems may be major contributing factors to the NO₂ levels in this region. The southeast and eastern parts show markedly lower NO₂ levels. This area is mostly marked by extensive cropland, indicating higher agricultural activity rather than industrial production. The extreme northern and northeastern border regions also show lower NO₂ levels which is in line with the region being forest land.

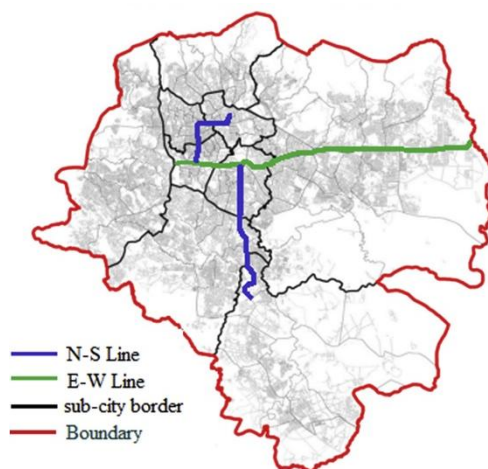


Fig. 4.2 Addis Ababa LRT Map
Credit: Teklemariam and Shen (2020)

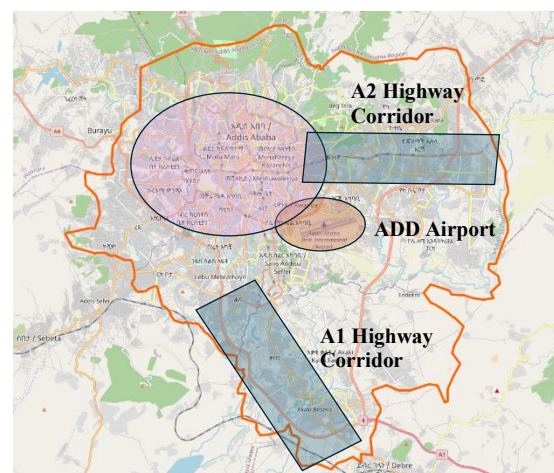


Fig. 4.3 Addis Ababa Road Network and Airport
Credit: OpenStreetMap (2025)

On closer inspection of Figure 4.1, we also notice a variation in NO₂ levels on weekends as compared to weekdays. This may be associated with lower economic activity on the weekends such as less people travelling to work and school, businesses being shut, and increased home-based activities.

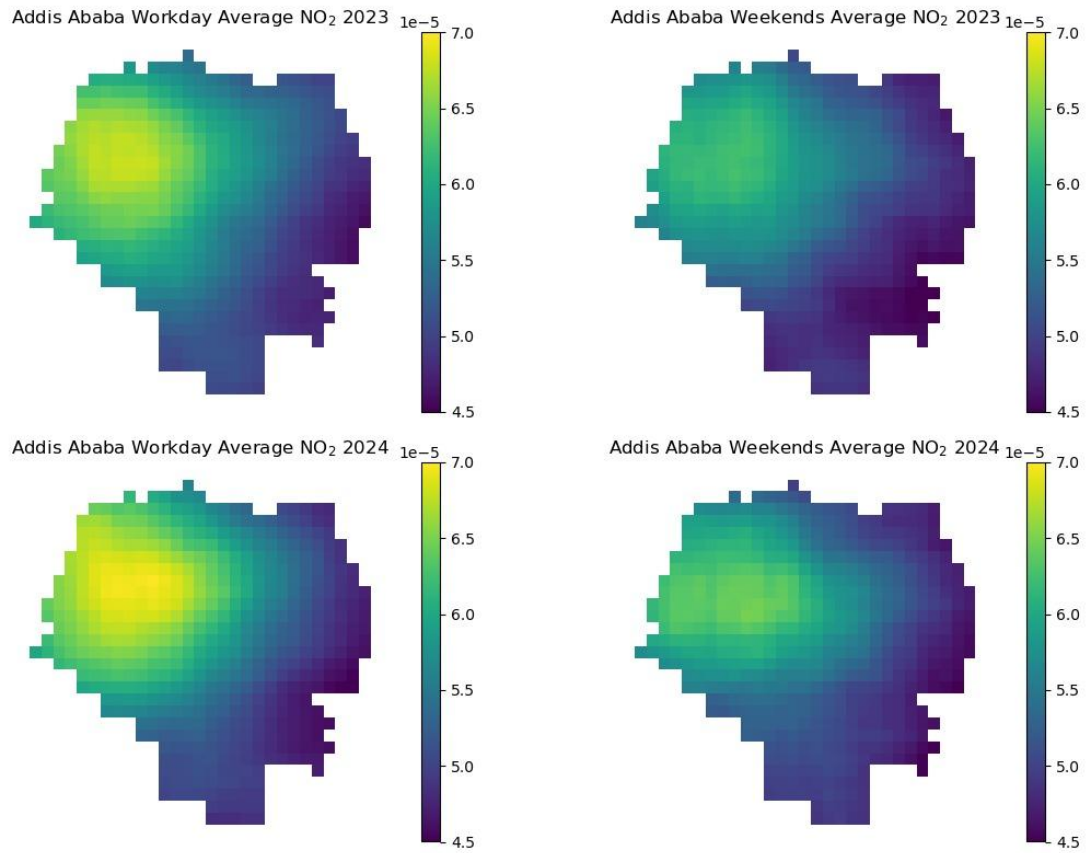


Fig. 4.4 Addis Ababa NO₂ levels on Weekdays and Weekends in 2023 and 2024

A comparison is made for NO₂ levels in 2023 and 2024 in Figure 4.4. As expected, weekday NO₂ levels remain higher due to commuter traffic, especially in the northwest core region. On weekends, the NO₂ levels are lower with a smoother spatial distribution. The NO₂ level for both workdays and weekends rose in 2024, this is likely from expanding road networks and vehicle numbers. Weekend levels have also increased, suggesting growing leisure, delivery, and industrial activities outside workdays due to the expansion.

The daily subregional NO₂ levels have been attached in Appendix Figure 3. All the subregions consistently show a spike in NO₂ levels in February-March and November-December, with a marked decrease in August. In Ethiopia, February-March and November-December fall in the dry season; with dry, stable atmospheric conditions and low wind which can trap pollutants near the surface due to poor vertical mixing. The period also coincides with pre-holiday commerce and traffic activity due to festivals and the peak tourist season. On the other hand, August sees the highest rainfall in Ethiopia, which may be the reason for the drop in NO₂ levels due to washing away of NO₂, as highlighted in section 2.2.

The NO₂ levels in the city are also analysed using autocorrelation. Temporal autocorrelation and spatial correlation are computed to investigate the correlation of NO₂ with itself over time and

over the regions of interest. The NO₂ levels in Addis Ababa exhibit short-term autocorrelation, supporting the use of low-order autoregressive models for prediction.

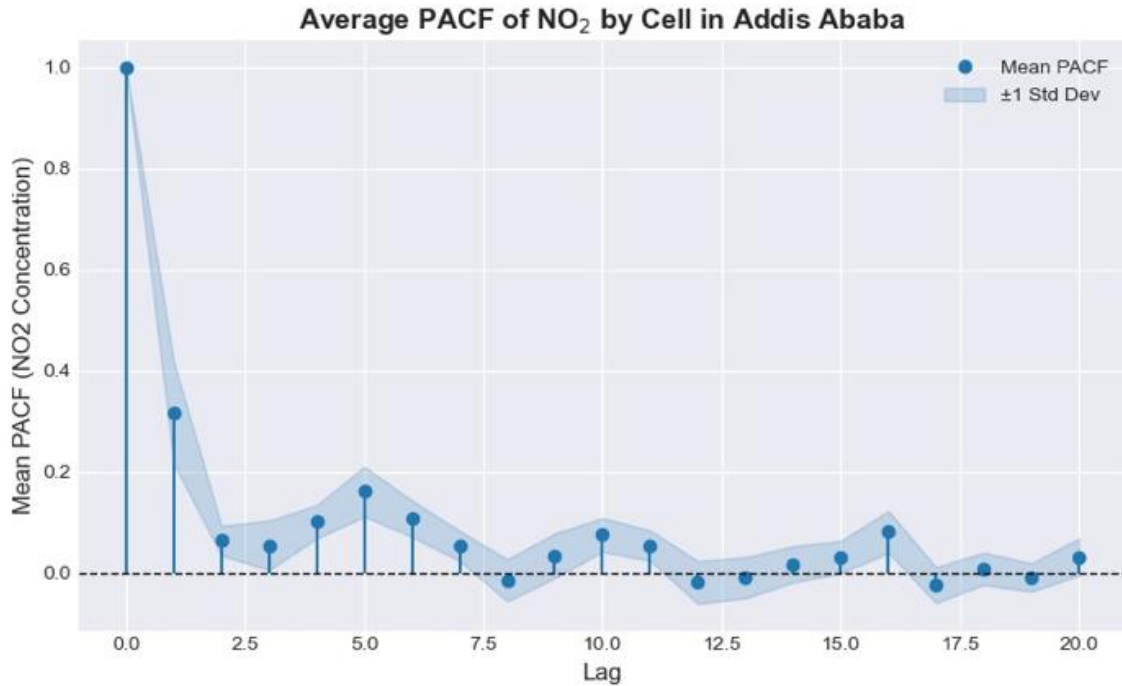


Fig. 4.5 Addis Ababa Temporal Autocorrelation

As evidenced by Figure 4.5, there is a strong positive partial autocorrelation at lag 1, indicating significant short-term persistence, but the influence drops off rapidly after a lag of one day. This is in line with NO₂'s short lifetime of less than a day. Minor peaks at lags 5, 10, and 15 suggest possible weekly traffic cycles, while the confidence interval (± 1 standard deviation of PACF values across all cells) reflects spatial heterogeneity across grid cells.

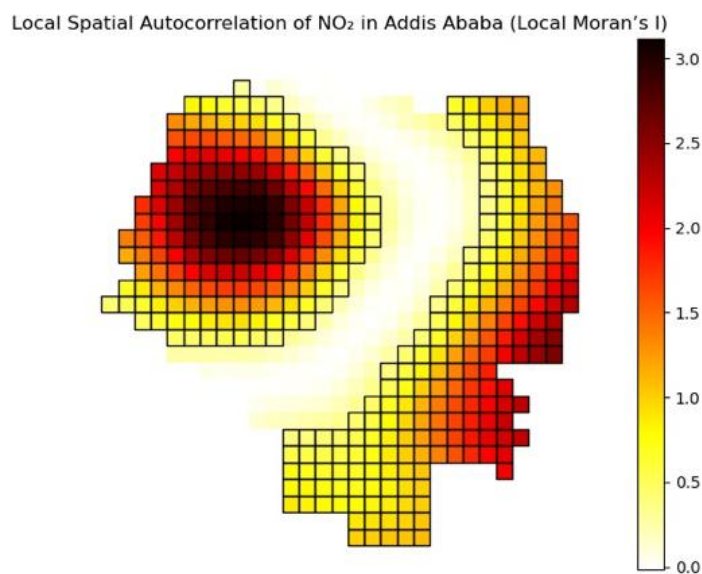


Fig. 4.6 Addis Ababa Spatial Autocorrelation

The spatial correlation in Figure 4.6 shows the correlation of NO₂ level with its neighbouring locations over two years between 2023 and 2024. The pollution is spatially dependent and concentrated, where high local spatial autocorrelation is concentrated in the western and central areas. High Local Moran's I values cluster around the city centre and in the outer region, indicated by dark red and black zones. The high Local Moran's I value at the centre, around 3.0, indicates strong local clustering properties around the area. The spatial gradient of NO₂ clustering indicates the potential for local spillover effects, where pollution in one area affects nearby regions due to consistent localised resources, such as daily traffic emissions.

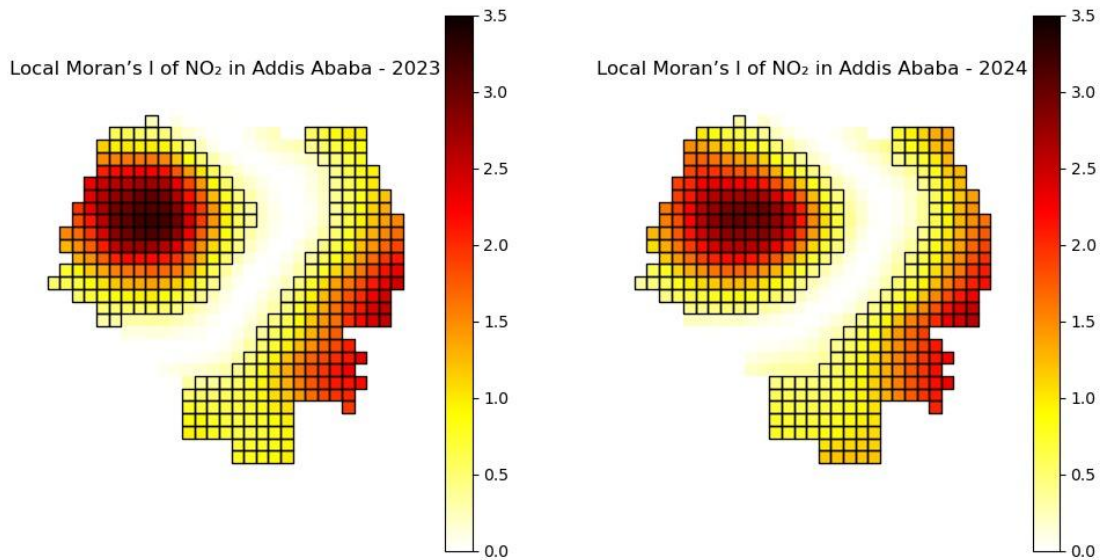


Fig. 4.7 Addis Ababa Spatial Autocorrelation (2023 v/s 2024)

In contrast to 2023, several regions in the northeast, southeast outskirts and eastern suburban zones transitioned from low/no autocorrelation (yellow/white) to moderate–high autocorrelation (orange/red) in 2024. This is shown by Figure 4.7. In 2023, the high-local Moran's I values are slightly more concentrated. On the other hand, in 2024, the high autocorrelation zone appears to diffuse slightly outward.

4.1.2 Population

The population distribution for 2023 and 2024 also shows a similar trend to NO₂, i.e., a possibly high correlation to the transport network and industrial production regions. Particularly, two offshoots are observed spreading from the northwest horizontally to the east and diagonally to the south. For the first case, the presence of the East-West line of the LRT corridor and A1 highway may be a possible reason. For the second case, higher population is observed along the North-South line of the LRT followed by the A1 highway corridor.

Similar to NO₂, lower population numbers are observed in croplands and forested regions. Between 2023 and 2024, the total population in Addis Ababa increased by 2.17% to about 4,281,917. The boxplot in Appendix Figure 1 shows a mean of 7336 in 2023 and 7842 in 2024. The histogram in Appendix Figure 2 shows that a considerable number of cells have population values below 5000.

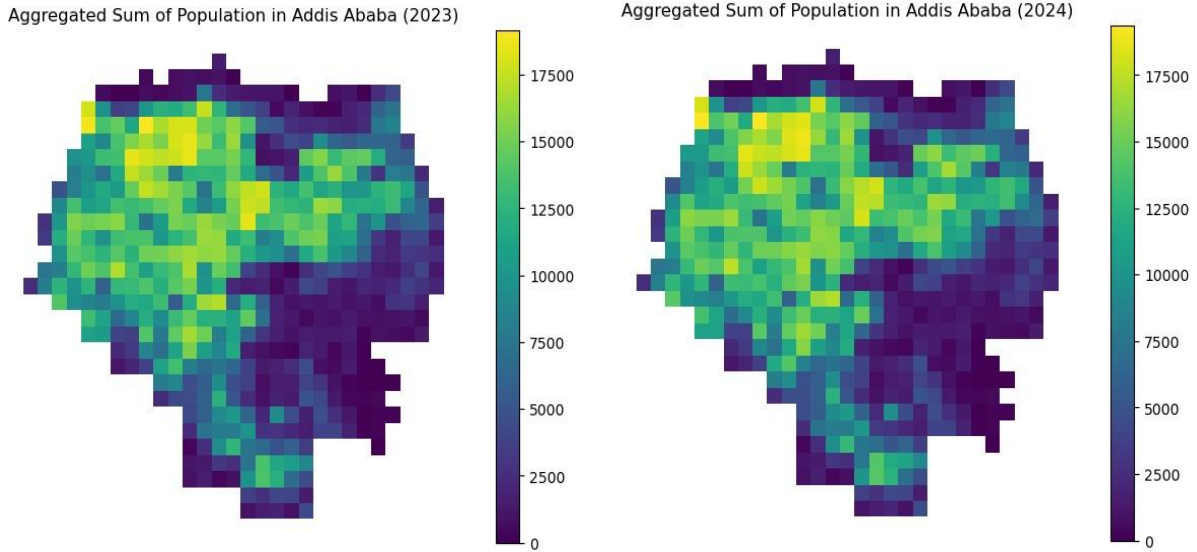
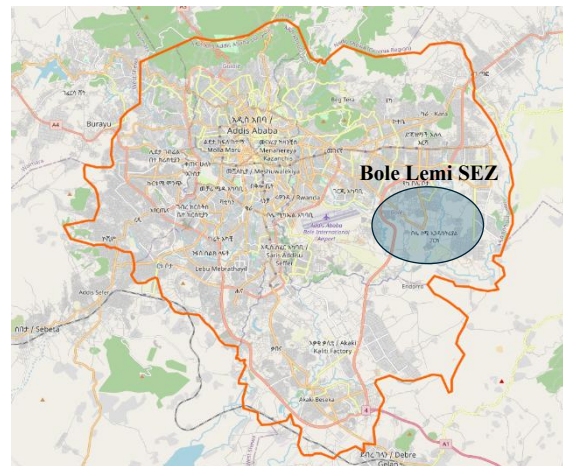
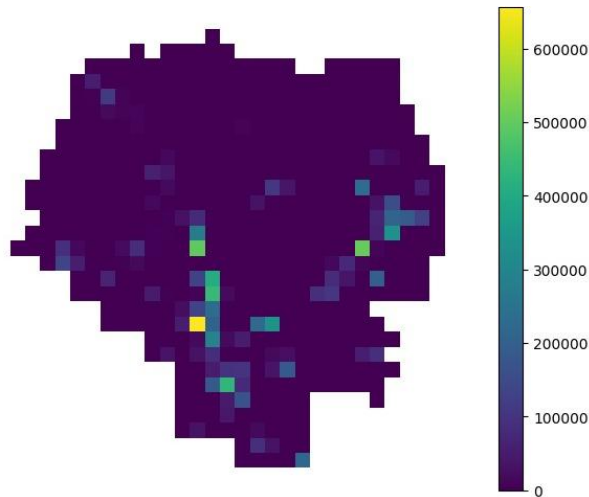


Fig. 4.8 Addis Ababa Aggregated Population per Cell

4.1.3 Land Use (Industrial)

The industrial land use area distribution as shown in Figure 4.9 also shows higher values along the A1 highway corridor. Furthermore, higher industrial area towards the east can be attributed to the Special Economic Zone (SEZ) of Bole Lemi as shown in Figure 4.10.



4.1.4 Points of Interest

The points of interest data was processed to obtain the total percentage of POI expected to produce considerable amounts of NO₂ due to industrial activity and/or mobility. These POI included supermarkets, hospitals, car dealerships, car rentals, carwash, marketplaces, malls, and universities. Comparing Figures 4.3 and 4.11, a higher concentration of POI is observed in the heart of Addis Ababa.

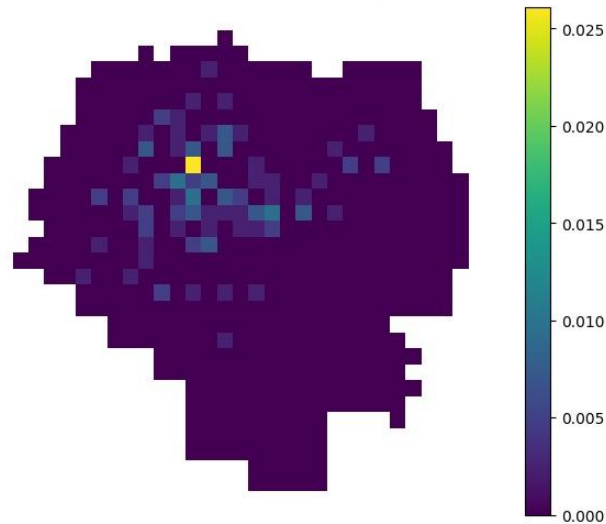


Fig. 4.11 Addis Ababa Points of Interest (in %) (2023)

4.1.5 Road Length

The total road length per cell was calculated in metres per cell as shown in Figure 4.12. Comparing this with Figures 4.2, 4.3, and 4.8, we observe a correlation between the persistence of roads, LRT, and population distribution.

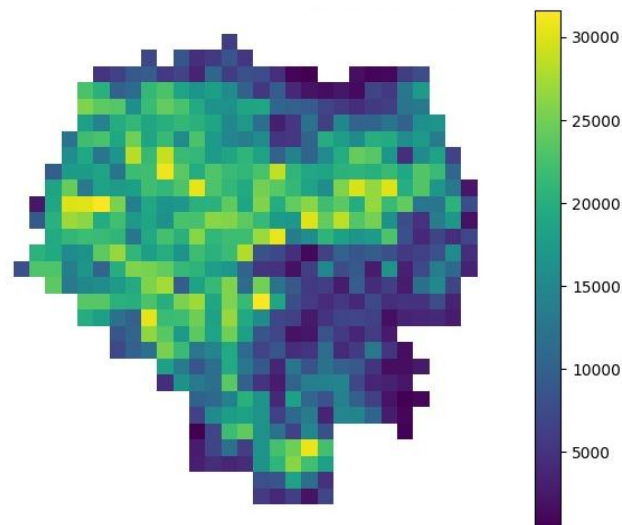


Fig. 4.12 Addis Ababa Total Road Length (in m) (2023)

4.1.6 Distance to Primary Road

To investigate the effects of distance from the closest primary road, the cell with the primary road is categorised as 1, while the ones that do not contain primary roads are categorised as 0.

Figure 4.13 shows that the cells in the northwest part of Addis Ababa have the most access to the primary road. The ones in the southeast part of the city, in Akaki Kality, tend to have a larger distance (around 6000 m) from the nearest primary road.

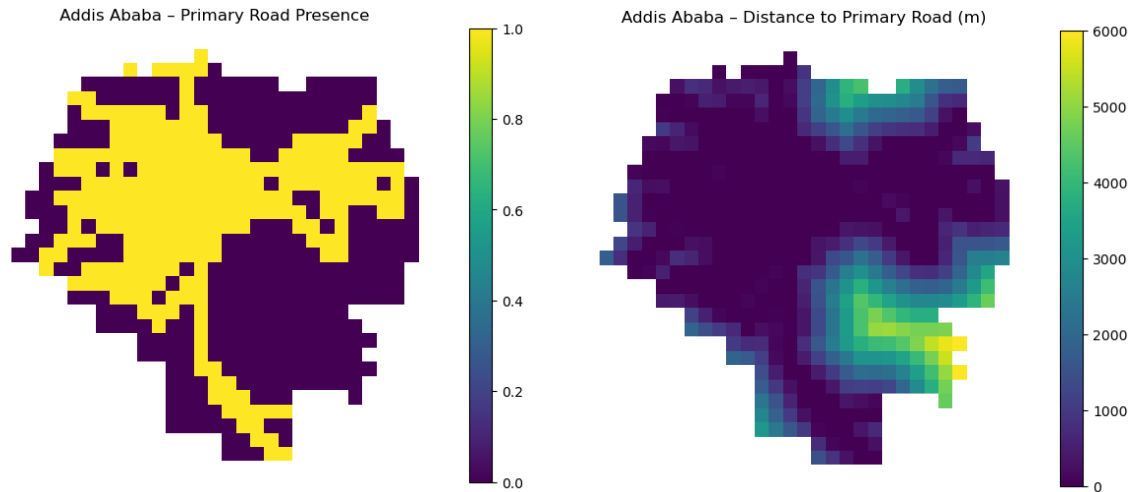


Fig. 4.13 Primary Roads and Distance to Primary Roads (Addis Ababa)

4.1.7 Land Surface Temperature

The maps in Figures 4.14 and 4.15 illustrate the spatial distribution of average land surface temperature (LST) in Addis Ababa for the years 2023 and 2024, derived from MODIS Terra (measured at 10:30 AM) and Aqua (measured at 1:30 PM) satellites. As expected, Aqua-based maps (Figure 4.14) show higher temperatures than Terra (Figure 4.15), due to increased solar heating by early afternoon.

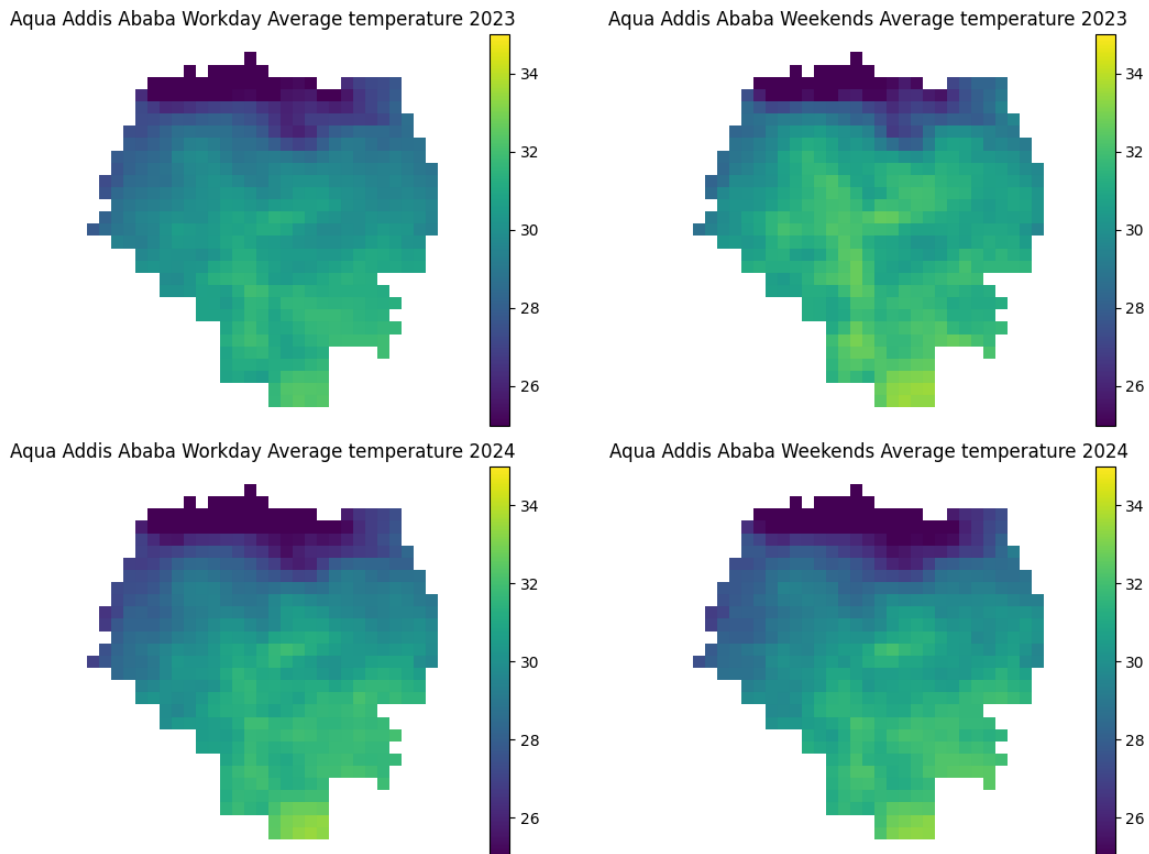


Fig. 4.14 Addis Ababa Temperature Distribution from Aqua (in °C) (2023-2024)

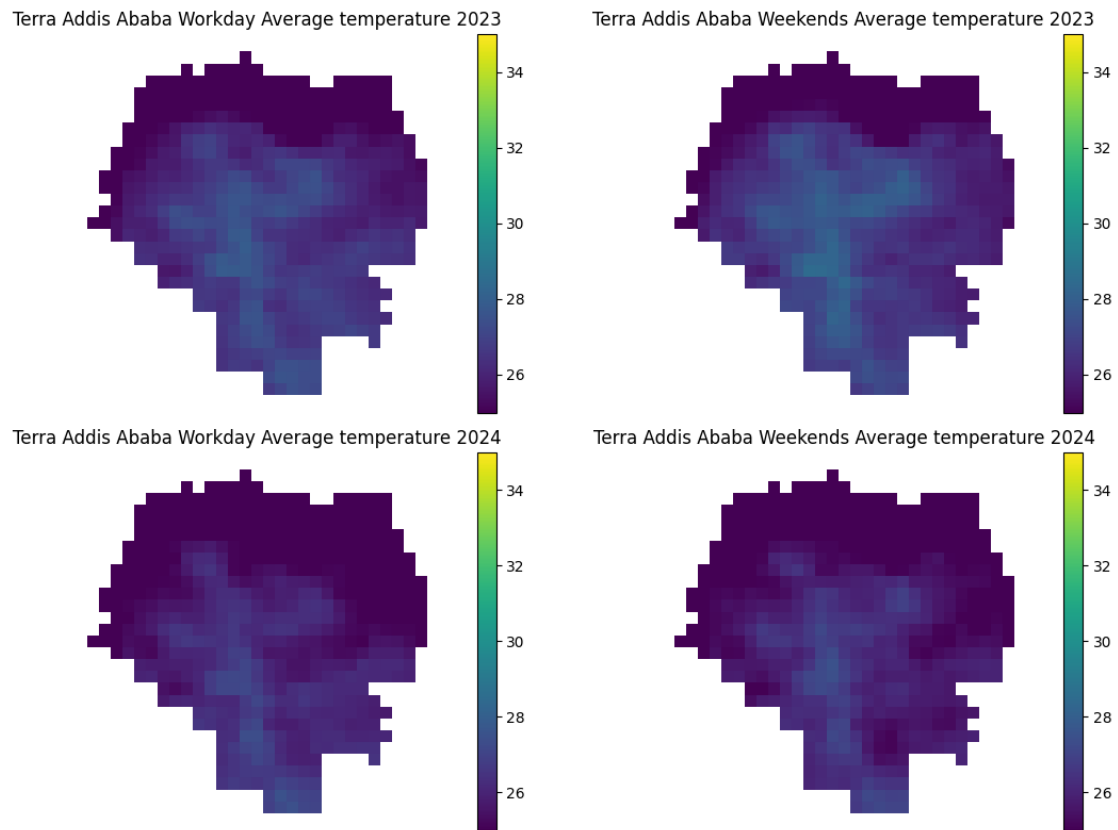


Fig. 4.15 Addis Ababa Temperature Distribution from Terra (in °C) (2023-2024)

A consistent pattern is observed across both satellite products: surface temperatures in 2024 are generally higher than in 2023, particularly during weekends. The southern and southeastern zones of the city consistently exhibit elevated temperatures, likely corresponding to lower-elevation and less vegetated areas. Meanwhile, the northern highland areas of Addis Ababa remain relatively cooler throughout all scenarios, possibly due to the existing tree cover.

A subtle yet noticeable difference appears between workday and weekend temperatures, especially in the Aqua maps. Weekend heating is more intense, particularly over the central region and peripheral urban districts along the highway corridors.

The observed increase in temperature distribution across the city is largely attributed to anthropogenic factors (McCarthy, 2001). Migration, political centralisation, and economic opportunities within the city have led to rapid urbanisation and population growth, resulting in a decrease in vegetation and green spaces alongside increased human and vehicular activity, including commercial and transportation emissions. The high congestion in the city centre reflects the key urban challenges facing Ethiopia, where a combination of inadequate public transport systems, insufficient infrastructure for non-motorised transport, and unplanned spatial expansion causes the traffic congestion (UN-Habitat, 2025).

4.1.8 Cloud Cover Category

As illustrated in Figure 4.17, in both years, the southern and southeastern parts of the city consistently experience clearer skies, while the northern and central zones tend to be cloudier.

Cloud cover towards the north is influenced by a combination of factors related to the city's high altitude and the surrounding topography. The interaction of warm, moist air rising over the Ethiopian highlands can lead to condensation and cloud formation as warm, humid air is forced upwards by the terrain. The hills and mountains in the northern part of the city create a natural barrier for airflows, forcing warm, moist air upwards, causing it to cool and condense into clouds (Figure 4.16).

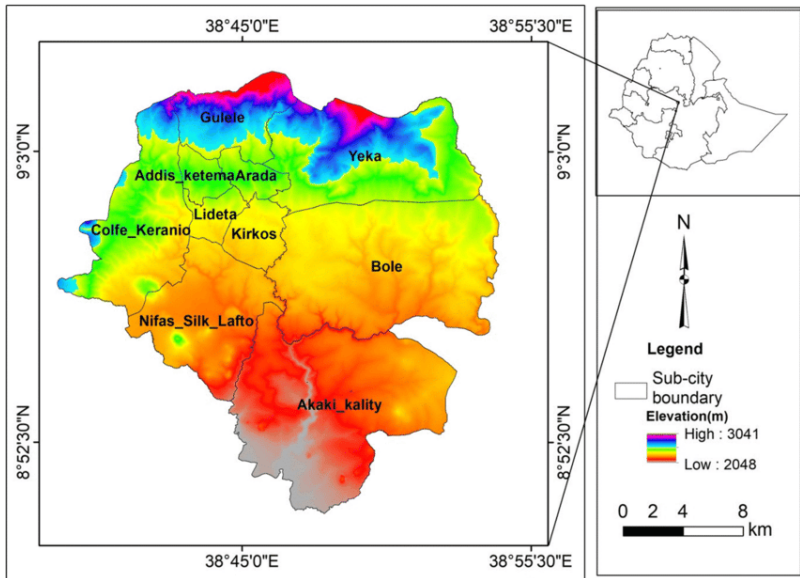


Fig. 4.16 Evaluation Map of Addis Ababa
Credit: Feyissa et al. (2018)

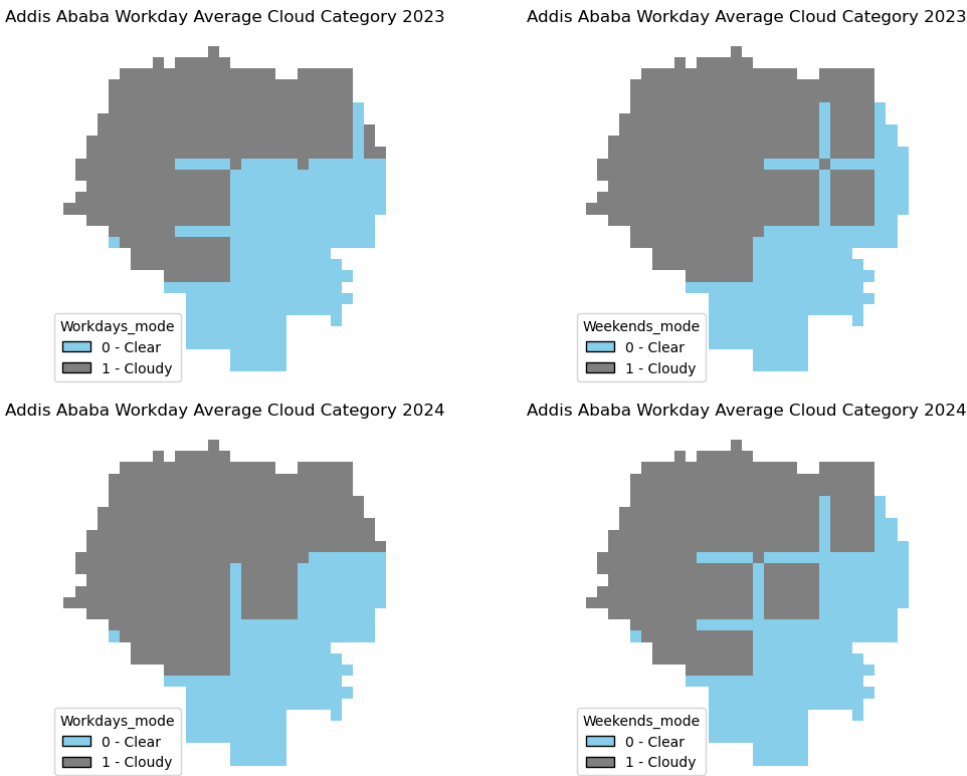


Fig. 4.17 Addis Ababa Cloud Cover Category (2023-2024)

4.1.9 Land Cover (ESA)

As shown in Figure 4.18, most built-up regions are centred around the central and northeast part of the city, and the southeast part of the region is mostly cropland. Tree cover, indicated in dark green, is mainly concentrated in the north and scattered across the central-west areas, while the grassland is spread around the city.

Shrubland and grassland are primarily found in less densely developed urban areas, and bare or sparsely vegetated zones are located along the southern and eastern edges of the city. The areas of water bodies and wetlands are limited, with only small and isolated patches of blue and teal barely visible on the map.

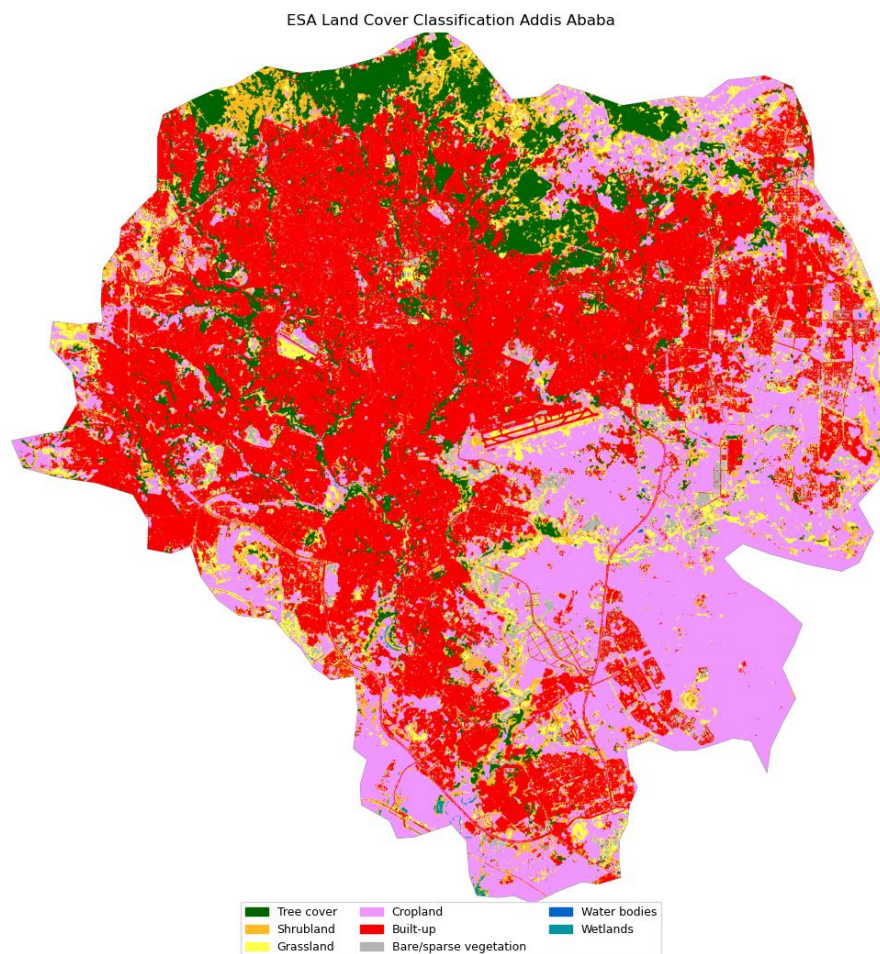


Fig. 4.18 Addis Ababa Land Cover (2023-2024)

4.2 Baghdad

4.2.1 NO₂

The weekend in Iraq is Friday-Saturday. For the purpose of subsequent traffic congestion analysis, the week has been divided into three parts: workday (Monday-Wednesday), extended workday (Thursday, Sunday), and weekend (Friday-Saturday). As mentioned in section 2.1.2, the area under consideration is the Baghdad Governorate.

The spatial distribution of NO₂ concentrations in Baghdad for 2023 and 2024 highlights persistently elevated emissions in the urban core during workdays and extended workdays, correlating with traffic density and continuous industrial activity.

Workday averages consistently show higher NO₂ levels around the city centre, reflecting the increase in vehicular and factory operations. Workday emissions in 2024, as indicated in Figure 4.19, contrasted with a modest increase in weekend pollutant levels. The increasing central-city pollution is mostly attributed to ongoing activities in vehicular traffic and industrial facilities, reinforcing the effects of daily urban mobility and extended working hours on NO₂ pollution.

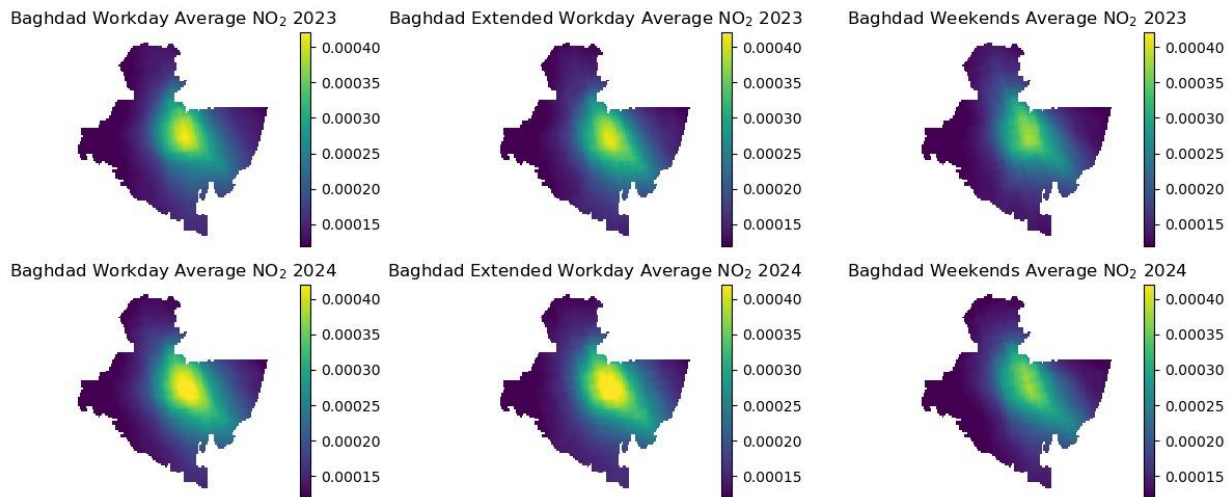


Fig. 4.19 Baghdad Average NO₂ Levels 2023-24 (in mole/m²)

As highlighted in Figure 4.20, the central region is the city of Baghdad. It is the capital and largest city of Iraq and thus, the nation's economic hub. The green spot in the figure indicates Baghdad International Airport (BGW). As shown in Figure 4.21, moving towards the south from the city centre is a major mobility corridor formed by the Baghdad–Basra Railway Line, and Highway 8 and Motorway 1 (M1) which connect Baghdad to Kuwait. The figure also shows Highway 6 which goes from the city centre towards the southeast to Al Kut and connects Baghdad to Tehran.

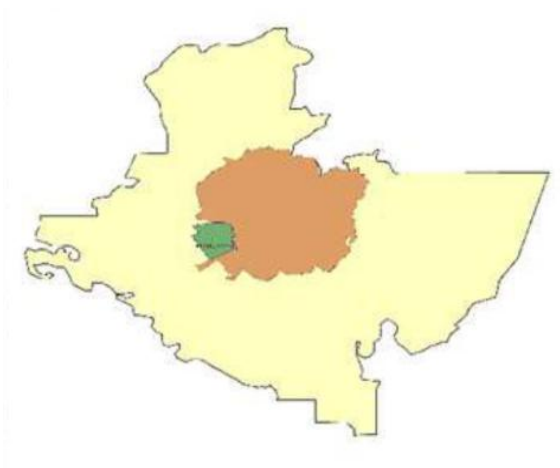


Fig. 4.20 Baghdad Province Map
Credit: Hassoon and Roomi (2023)

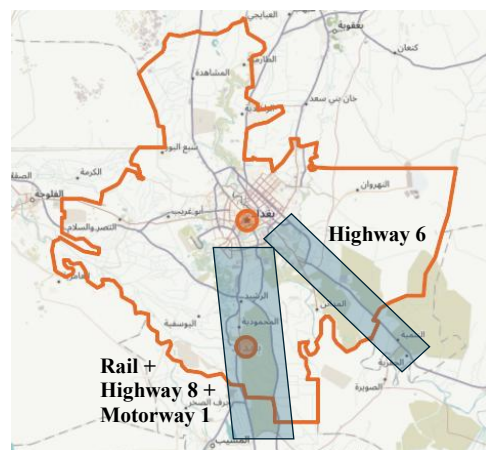


Fig. 4.21 Baghdad Road Network
Credit: OpenStreetMap (2025)

As shown in Figure 4.19, the highest NO₂ concentrations are observed in Baghdad city centre; where most of the urban activity is concentrated. Considerable emissions are also observed as we move towards the southeast, along the Highway 6 corridor. Similar to Addis Ababa, there is a slight variation in emissions for weekdays and weekends. The reasons for this variation may also be similar.

The daily subregional NO₂ levels attached in Appendix Figure 4 show that emission levels in the outer regions are fairly consistent throughout the year. On the other hand, for the central regions, the emissions profile is more peaky, with spikes often seen between July-September. These months are marked by high temperatures and clear skies along with power cuts. This leads to increased electricity demand for air conditioning and usage of diesel generators. The period also coincides with Muharram, which often involves communal events and processions drawing large crowds.

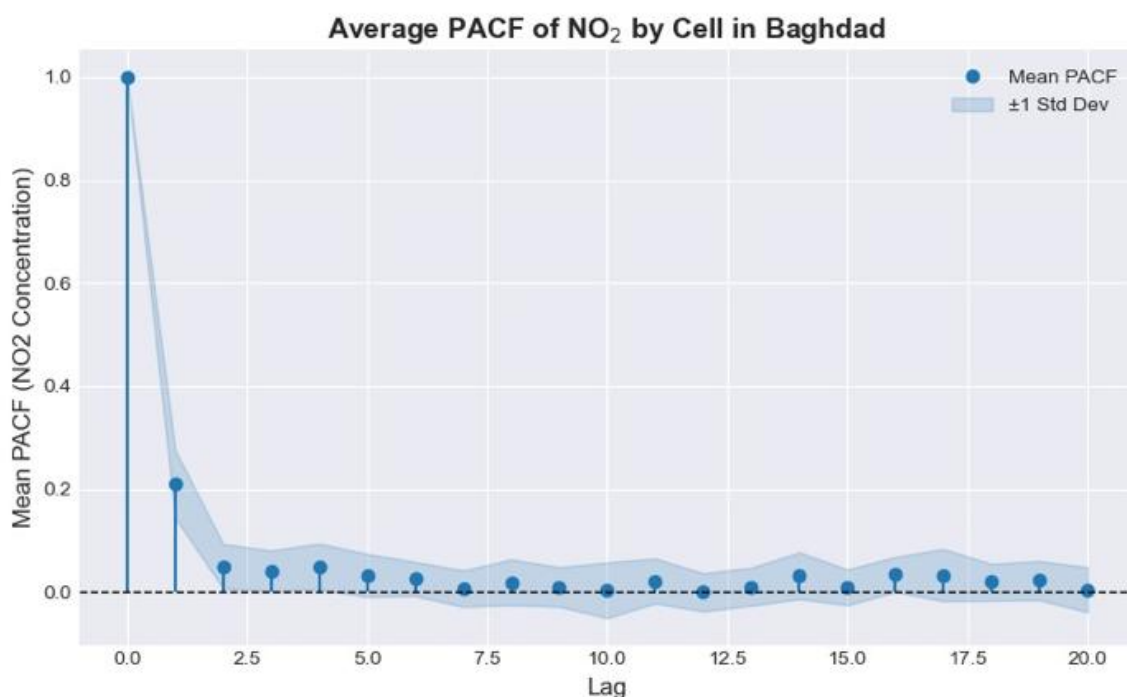


Fig. 4.22 Baghdad Temporal Autocorrelation

Similar to Addis Ababa, the NO₂ levels in Baghdad show short-term correlation. The average PACF value of Baghdad shows low autocorrelation at lag 1, with all subsequent lags hovering near zero. The narrower confidence interval in Baghdad suggests more homogeneous and random temporal behaviour. The smooth PACF at later lags imply the NO₂ concentrations may change independently from one day to the next, indicating that episodic factors, such as weather, likely influence NO₂ levels in Baghdad.

Figure 4.23 shows a much sharper and more concentrated hotspot in the spatial autocorrelation of Baghdad. Outside this zone, spatial autocorrelation is nearly zero. The local Moran's I value at the centre of Baghdad is much higher than that of Addis Ababa. With a value of 8 at the city centre, Baghdad has a very strong local hotspot compared to Addis Ababa, which has more moderate clustering spread over a larger area.

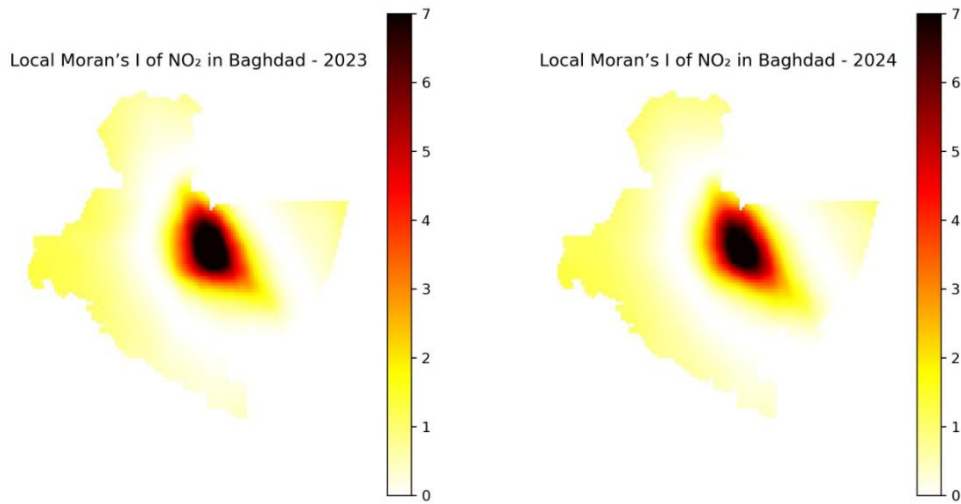


Fig. 4.23 Baghdad Spatial Autocorrelation in 2023 and 2024

Figure 4.23 above shows the highest autocorrelation in central-eastern Baghdad, with the highest local Moran's I value in the city centre being higher than 6.0. The orange/yellow zones (Local Moran's I $\approx$ 2–4) in the northwest and southwest parts of the city appear to have widened in 2024, indicating a stronger consistency in NO₂ concentrations. On the other hand, there is less extreme clustering within the center in 2024 compared to the previous year. This shift suggests that NO₂ pollution has become more spatially diffuse, likely due to the redistribution of traffic as well as meteorological factors (e.g., temperature inversions) that trap pollutants more broadly (iData, 2024).

4.2.2 Population

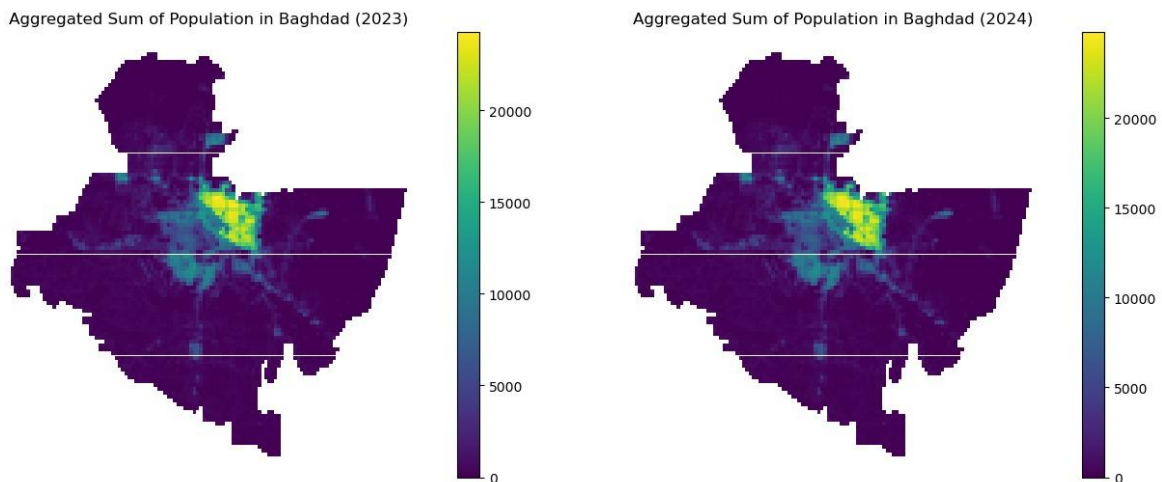


Fig. 4.24 Baghdad Aggregated Population per Cell

In Baghdad, the population distribution shows a similar profile to NO₂ emissions distribution, with a higher concentration in the city centre. The population is particularly high in Sadr City and Al Baladiyat to the north of the Army Canal, parallel to the Tigris River. Closer inspection shows considerable population concentration along the Tigris and Highway 6 to the southeast and along the rail line in the south.

The population distribution is much lower in the outer regions, which are mainly defined by rangeland and cropland. The population increased by 199,125 between 2023 and 2024, a 2.10% increase. The boxplot in Appendix Figure 1 shows a mean value of 1547 in 2023 and 1579 in 2024. However, a large number of cells lie beyond the range. This, coupled with the histogram in Appendix Figure 2, indicates a few cells have very large population values compared to others.

4.2.3 Industrial Area and Points of Interest

The obtained industrial area data shows sparseness, as does the points of interest data. The trend of higher human activity concentrations in the city centre and along the highway corridors continues to prevail.

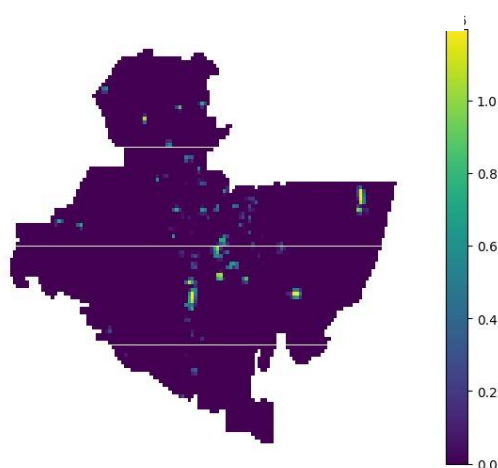


Fig. 4.25 Baghdad Industrial Land Use Area per Cell (in m²) (2023)

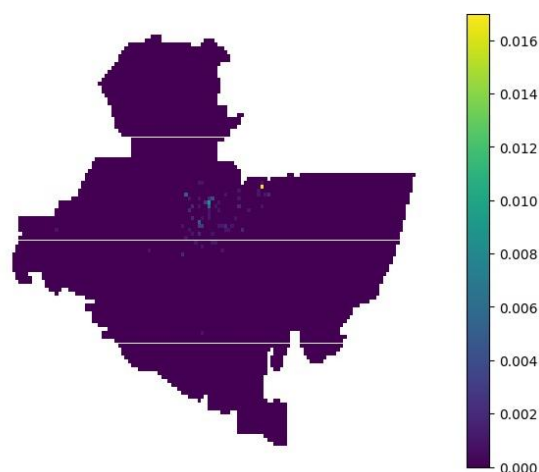


Fig. 4.26 Baghdad Points of Interest (in %) (2023)

4.2.4 Road Length

The total road length per cell in the grid shows a stark resemblance to the population distribution and NO₂ emissions plot. This shows a strong correlation between human settlement, mobility, and air pollution.

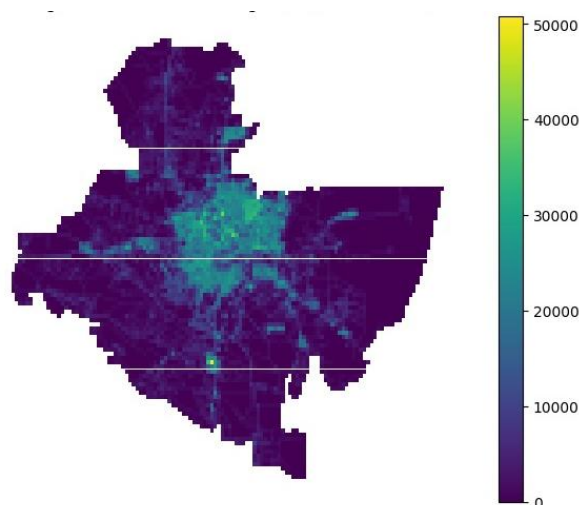


Fig. 4.27 Baghdad Total Road Length (in m) (2023)

4.2.5 Traffic Congestion Intensity (TCI)

Similar to NO₂, the average Traffic Congestion Intensity (TCI) in Baghdad is analysed across three temporal categories: regular workdays, extended workdays, and weekends.

On regular workdays, the congestion index is highest in the city centre and along major arterial roads, indicating significant commuting activity concentrated in core business zones. During extended workdays, the spatial extent of congestion expands slightly, suggesting the potential of longer working hours or late-day activities. In contrast, the congestion index is considerably lower on weekends in both intensity and spatial coverage, although the city centre still exhibits moderate levels of TCI.

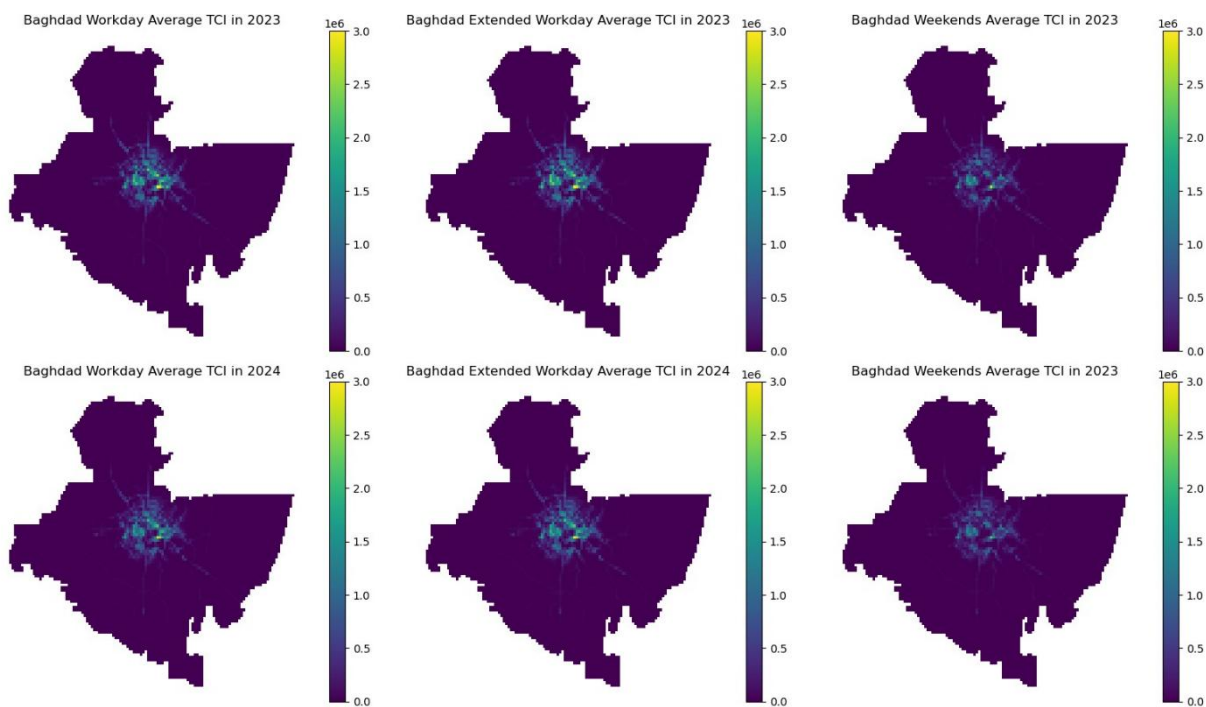


Fig. 4.28 Baghdad Traffic Congestion Intensity (2023-2024)

4.2.6 Land Surface Temperature

Similar to Addis Ababa, the spatial distributions of average land surface temperatures (LST) from MODIS Terra (10:30 AM overpass) and Aqua (1:30 PM overpass) satellites are considered. A clear seasonal and temporal gradient is visible for both satellites, with Figure 4.30 showing that the average temperature in 2024 is slightly higher across all day types.

Higher temperatures tend to concentrate in the city centre and industrial zones, while peripheral areas remain relatively cooler. Weekday maps display intensified heating in the western, central, and eastern Baghdad. Weekend maps show a modest temperature increase, particularly in the city centre, indicating the potential of enhanced anthropogenic heat emissions from commercial, traffic, and industrial activity in the city centre. The higher temperature in the eastern part of the city can be attributed to sparse vegetation.

The results of the temperature distribution within the city center are supported by Alshammari et al. (2025), who used Sentinel-3 satellite Land Surface Temperature (LST) data and OpenStreetMap urban infrastructure data to confirm a notable rise in temperatures of up to 1.3°C in Baghdad, attributed to the urban heat island effect.

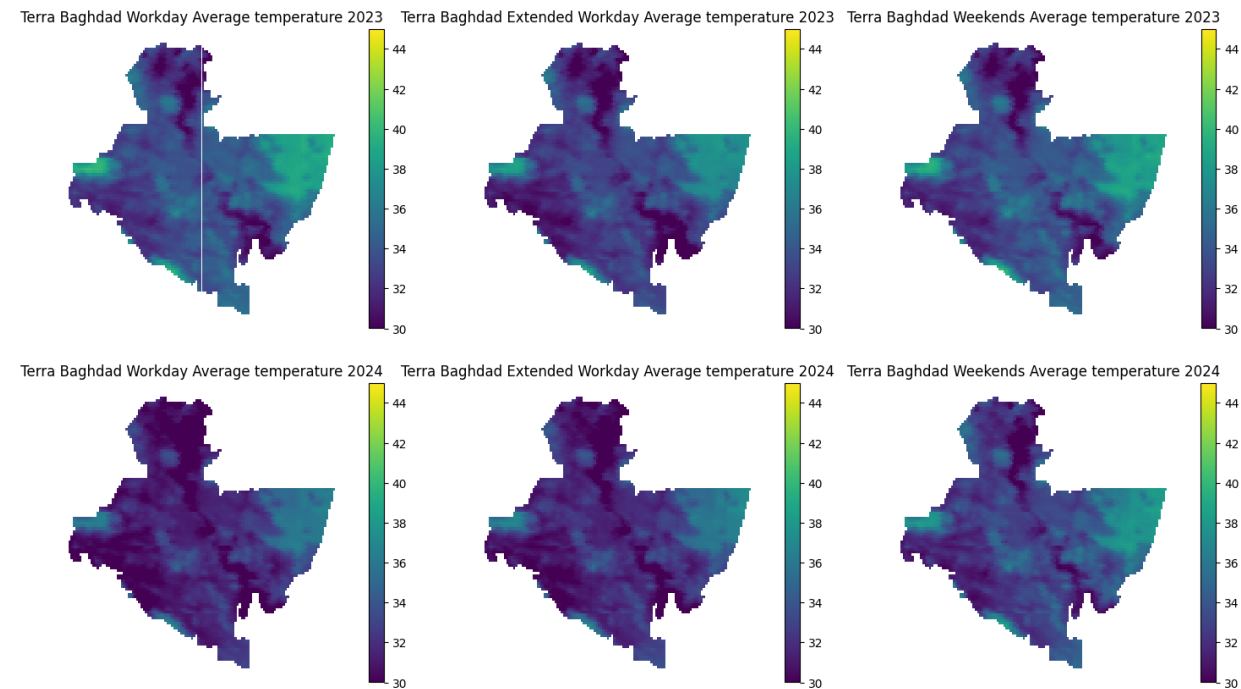


Fig. 4.29 Baghdad Temperature Distribution from Terra (in Degrees Celsius) (2023-2024)

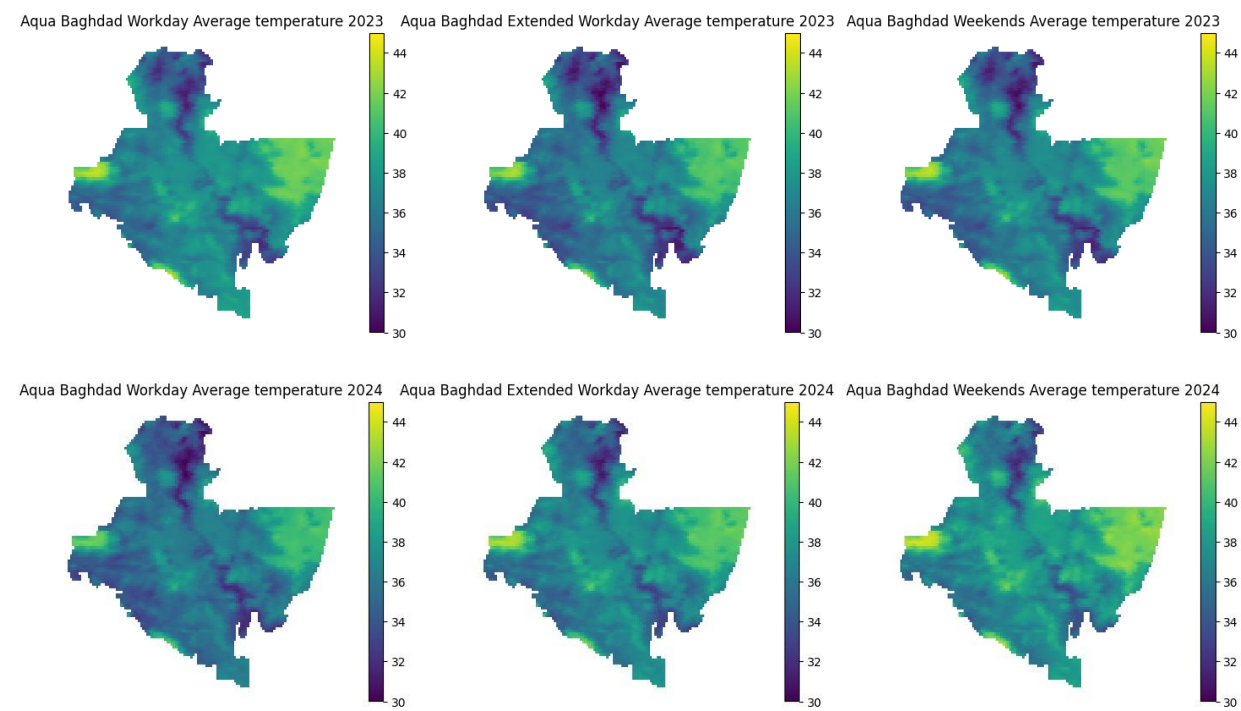


Fig. 4.30 Baghdad Temperature Distribution from Aqua (in Degrees Celsius) (2023-2024)

4.2.7 Cloud Cover Category

In this section, the most frequently occurring cloud cover category within each cell is analysed. Fig. 4.31 shows consistent clear-sky conditions (category 0) across all days of the week and the entire city for each year. The result reflects the weather condition in Baghdad, where the country experiences hot, dry summers characterised by clear skies. In winter, Baghdad sees some rainfall, although most of the year the sky remains clear. With an almost entirely flat and low-lying terrain, the city is less likely to form clouds.

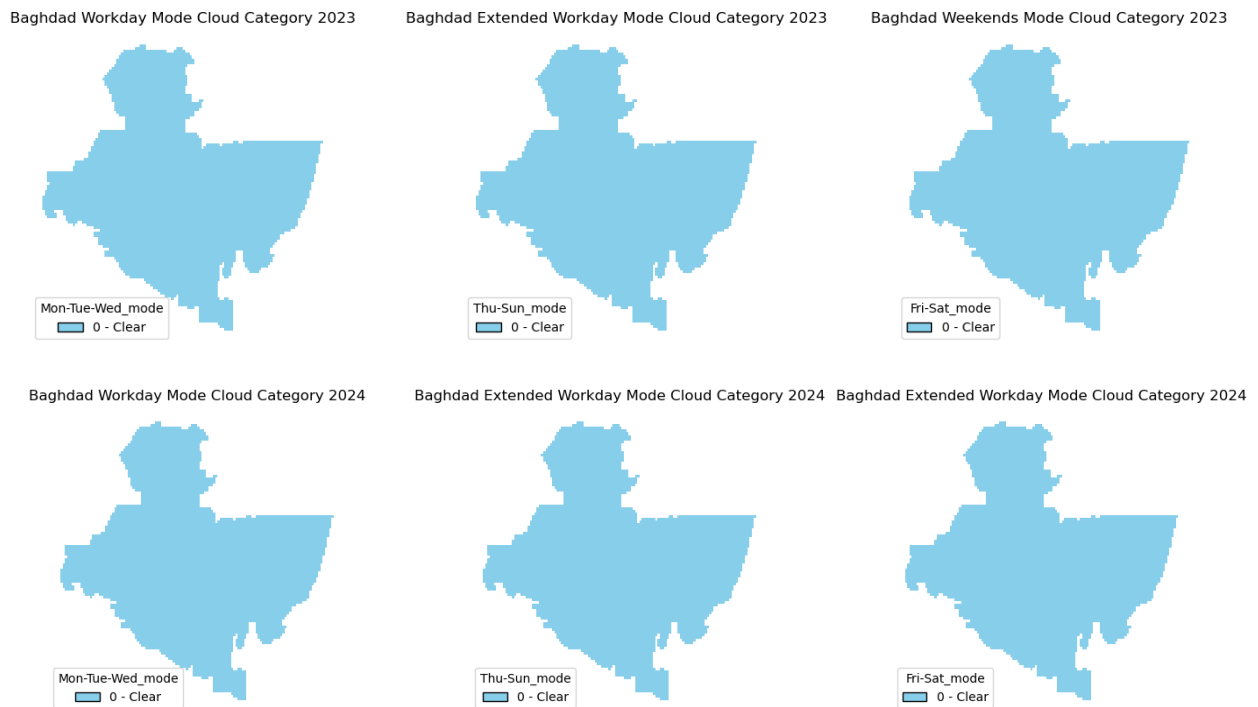


Fig. 4.31 Baghdad Cloud Cover Category (2023-2024)

4.2.8 Land Cover (ESA)

Figure 4.32 shows that the developed area in Baghdad is concentrated along the Tigris River and in the city's central region, where there are extensive vegetation and water-related land cover areas compared to Addis Ababa.

Extensive areas of tree cover and shrubland are particularly concentrated along the Tigris River and in the northern and eastern parts of the city. Cropland is widespread in all directions, especially east and south, which is consistent with the city's location within the Mesopotamian agricultural plain. Grassland and bare or sparsely vegetated areas are also prominent, especially toward the outer city limits. Wetlands are especially visible in the southern and eastern sections, which aligns with the seasonal flooding patterns and marshland ecosystems characteristic of the Tigris-Euphrates river basin.

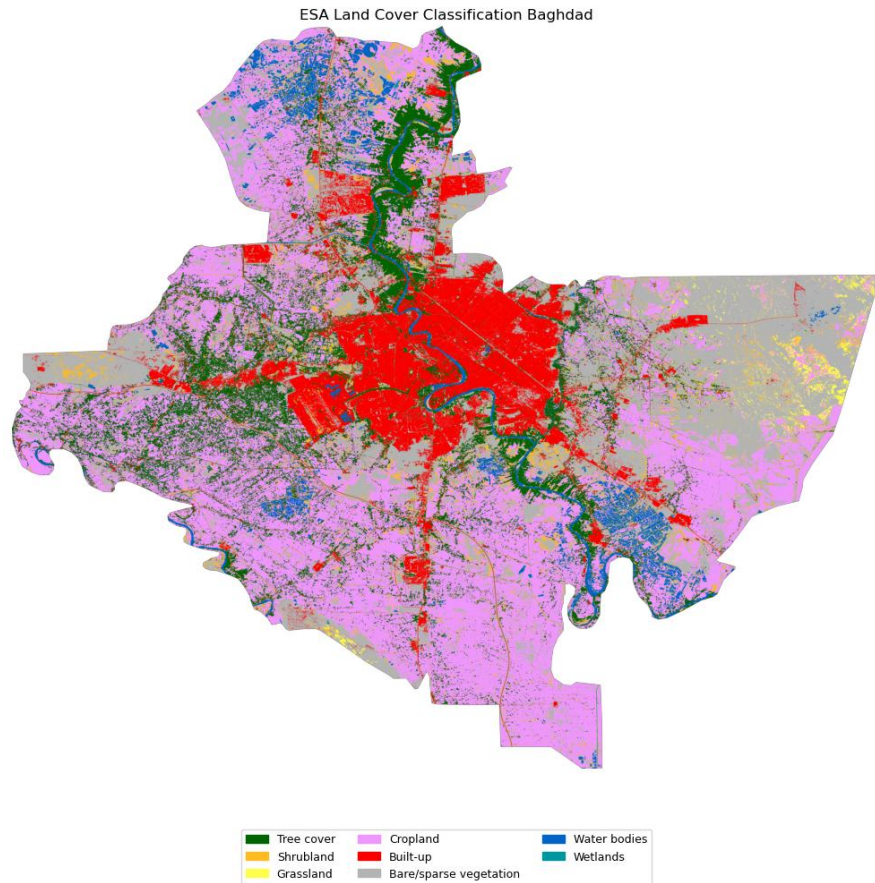


Fig. 4.32 Baghdad Land Cover (2023-2024)

4.3 Multivariate Correlation Analysis

To explore interdependencies between different features, correlation matrices were computed at the spatial grid-cell level. This analysis helps identify which factors are most strongly associated with NO_2 concentrations, providing insight into potential drivers of urban air pollution.

We calculated pairwise Pearson correlation coefficients among the selected variables. The correlation matrix quantifies linear relationships, where values range from -1 (perfect negative) to +1 (perfect positive). Strong positive correlations suggest shared spatial patterns or common underlying causes. Weak or negative correlations may point to inverse relationships or independent dynamics.

The resulting matrix is visualised using heatmaps to facilitate interpretation. This allows us to:

1. Identify variables with the strongest associations to NO_2 .
2. Detect multicollinearity, which informs feature selection in modeling.
3. Understand cross-variable interactions that shape pollution distribution.

This correlation analysis supports subsequent explanatory modeling and helps prioritise key predictors for intervention or policy consideration. The results are attached in the Appendix as Figures 5 and 6.

4.4 Air Pollutant Analysis

An analysis of different air pollutants, namely NO₂, SO₂, CO, and ozone, was done for both cities in 2023 and 2024. This was done to ascertain consistency between air pollutant trends.

4.4.1 Air Pollution Level in Addis Ababa

The correlation matrix for air pollution levels in Addis Ababa reveals significant interdependencies among key atmospheric pollutants. Notably, nitrogen dioxide shows a strong negative correlation with sulfur dioxide (SO₂) ($r = -0.71$) and ozone (O₃) ($r = -0.62$), indicating that periods of elevated NO₂ concentrations are typically associated with decreased levels of these two pollutants. Conversely, NO₂ exhibits a moderate positive correlation with carbon monoxide (CO) ($r = 0.59$), suggesting shared anthropogenic sources such as vehicular emissions.

SO₂ and O₃ are positively correlated ($r = 0.77$), as are SO₂ and CO ($r = -0.71$), although the latter indicates a potential source divergence. The strongest negative correlation in the matrix is observed between CO and O₃ ($r = -0.72$), suggesting the hypothesis of contrasting seasonal behaviours and atmospheric dynamics between primary and secondary pollutants.

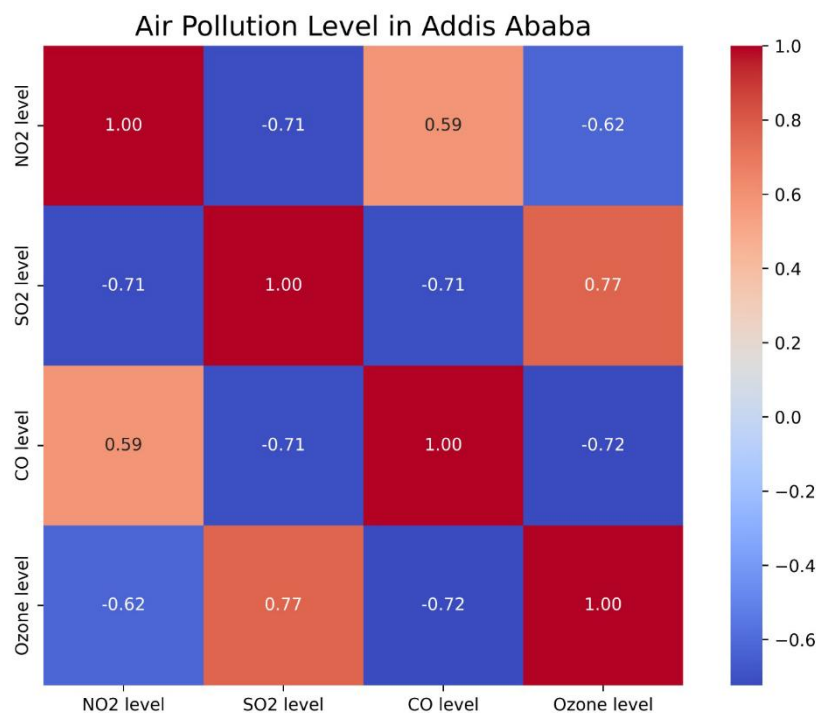


Fig. 4.33 Correlation of Air Pollutants in Addis Ababa

4.4.2 Temporal Trend of Air Pollution in Addis Ababa

The temporal trend of air pollution in Addis Ababa demonstrates variations across different pollutants, as illustrated in Figure 4.34. According to the result of smoothed moving average modelling, nitrogen dioxide and sulfur dioxide levels exhibit relatively high variability throughout the year. NO₂ peaks in early 2024 and late 2023, indicating the possibility of heightened vehicular or industrial activity during those periods, this aligns with the tourist season in Addis Ababa.

Conversely, SO₂ shows a major rise during the late months of 2023, followed by a decline toward the year's end. Carbon monoxide displays a pronounced seasonal cycle, reaching its highest concentrations around March 2023. Ozone levels present an inverse pattern compared to other pollutants, peaking around June and July, when photochemical reactions are enhanced by higher solar radiation.

The SO₂ concentrations exhibit periods of partial synchrony with NO₂ concentration, particularly during early 2023 and early 2024, suggesting overlapping emission sources such as vehicular traffic and industrial combustion. However, this relationship is not consistent year-round. Ozone concentration, in contrast, displays a distinct inverse relationship with NO₂ concentrations, with O₃ concentrations peaking during the summer months when NO₂ levels are at their lowest. This pattern reflects the photochemical processes in which NO₂ participates in ozone formation but is also depleted by it, especially under high solar radiation. Carbon monoxide concentration generally trends with that of NO₂ during colder months, likely due to common sources such as incomplete combustion from heating and transport.

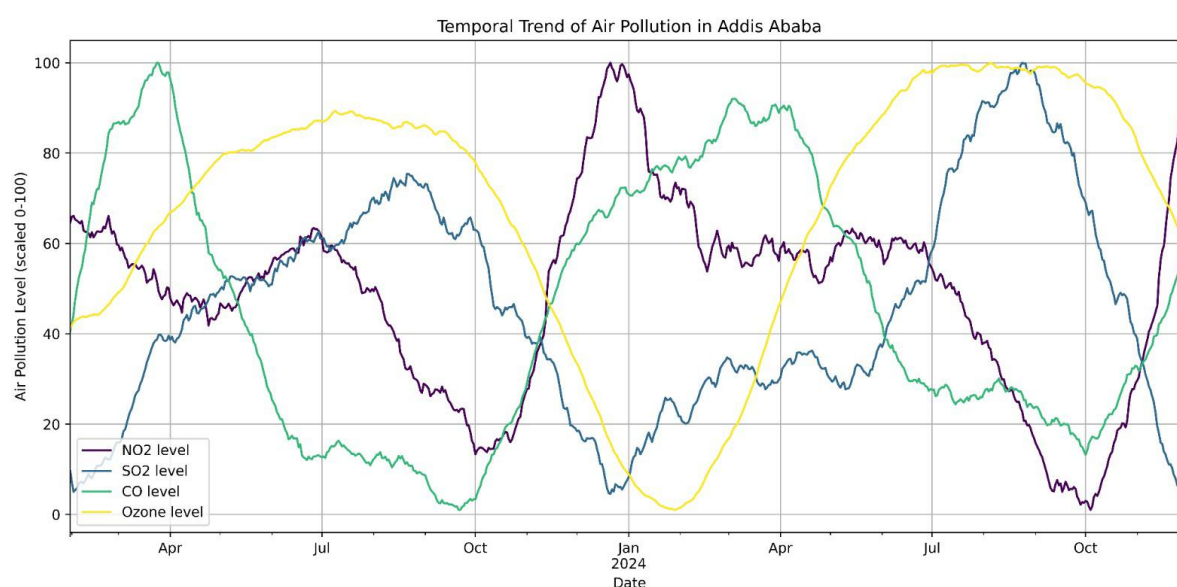


Fig. 4.34 Air Pollutant Trend in Addis Ababa

4.4.3 Air Pollution Level in Baghdad

The correlation matrix of air pollution levels in Baghdad presents a different interaction profile compared to that of Addis Ababa, suggesting distinct atmospheric dynamics and emission patterns.

Nitrogen dioxide displays a weak positive correlation with sulfur dioxide ($r = 0.27$) and carbon monoxide ($r = 0.25$), indicating some degree of co-emission, likely from transport and combustion sources such as diesel generators. Ozone, however, remains negatively correlated with NO₂ ($r = -0.53$), consistent with the well-documented photochemical relationship in which NO₂ is consumed during ozone formation. SO₂ however exhibits negligible correlation with CO ($r = 0.07$), suggesting they originate from different sources or are influenced by divergent meteorological factors.

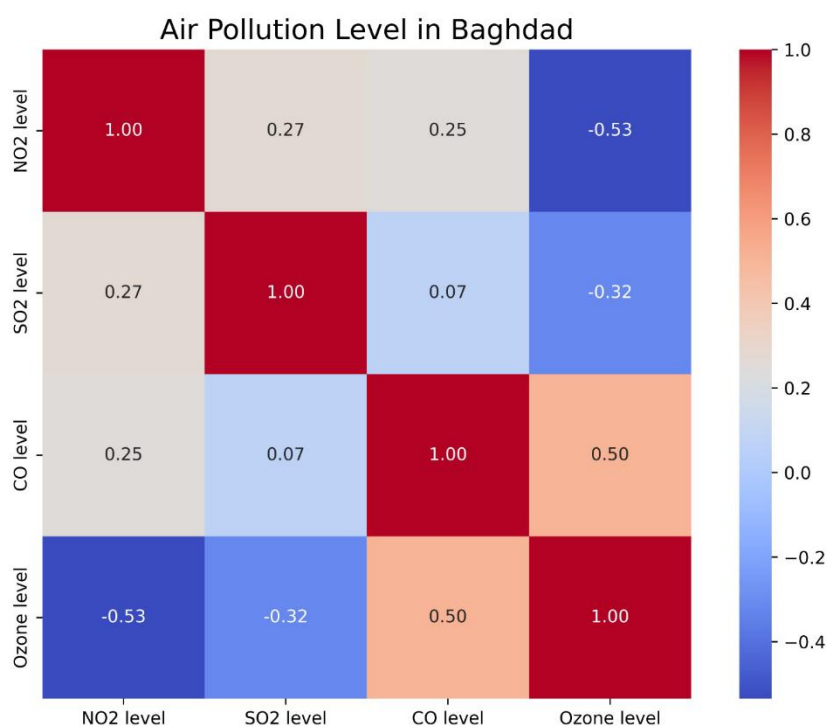


Fig. 4.35 Correlation of Air Pollutants in Baghdad

4.4.4 Temporal Trend of Air Pollution in Baghdad

According to smoothed moving average modelling, the temporal dynamics of air pollution in Baghdad reveal distinctive seasonal relationships between nitrogen dioxide and the other pollutants. SO₂ exhibits a strong positive correlation with NO₂ during the first half of 2023, as both pollutants peak concurrently in summer and decline afterward, suggesting shared emission sources such as traffic and industrial activity. However, this alignment weakens toward late 2023 and early 2024.

The carbon monoxide concentration shows a moderately synchronised trend with NO₂ concentration in the latter part of 2023 and early 2024, particularly during periods of heightened winter emissions, reinforcing the likelihood of overlapping anthropogenic sources such as vehicular exhaust and heating-related combustion.

In contrast, the ozone concentration demonstrates a strong inverse relationship with NO₂, particularly during the summer. This decrease in ozone may have been due to an increase in NO_x produced by diesel generators during heat waves, which suppressed the production of ozone. The inverse trend in April, just before summer, may have resulted from the photochemical processes in which sunlight-driven ozone formation consumes NO₂, especially under intense solar radiation.

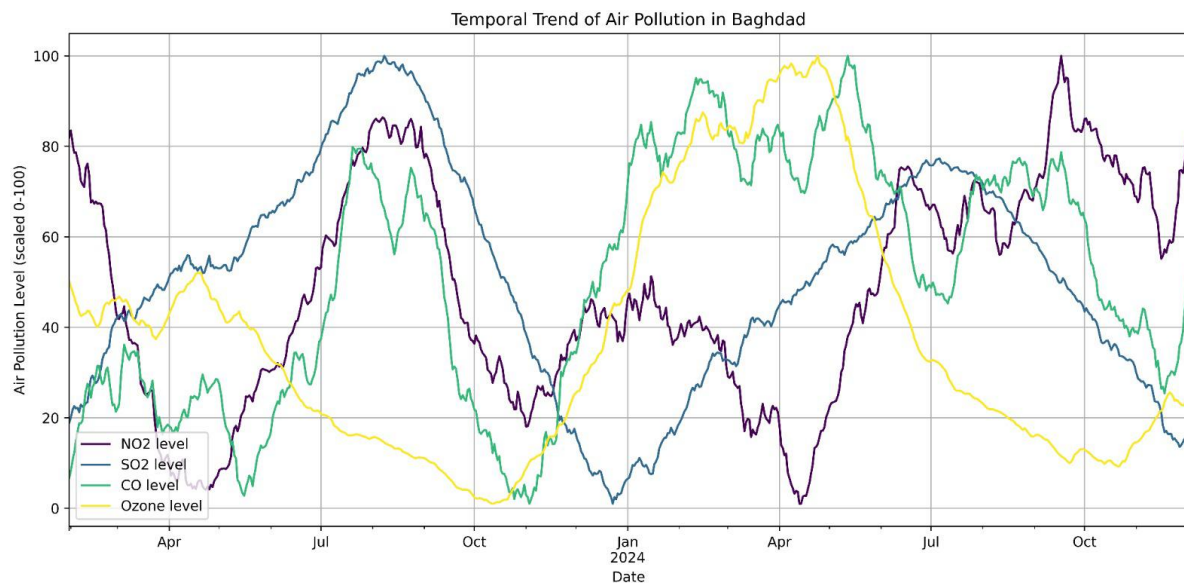


Fig. 4.36 Air Pollutant Trend in Baghdad

5. Methodology

The aim of this project is to run the following three models in order to address the research objectives:

- M1. NO₂ Model: Model NO₂ as a function of relevant explanatory variables including lagged variables and climate.
- M2. Industrial Production Model for Ethiopia: Model industrial production as a function of NO₂ and other relevant variables.
- M3. Mobility Model for Baghdad: Model congestion (i.e. Traffic Congestion Intensity) as a function of NO₂ and other relevant variables.

Through these three models, the applicability of NO₂ as a proxy for industrial production and mobility will be assessed. In addition, the possibility of attributing NO₂ to its contributors, such as weather and land use factors, will also be tested. Thus, research objectives 1, 2, and 3 will be directly addressed using these models.

5.1 R² Benchmark

For models that output an R² value, it is important to know if the value is acceptable. Based on available relevant literature, a short list of acceptable R² values (minimum 0.1) is made as shown in Table 5.8.

Table 5.1 Acceptable R² Values in Modelling

Authors	Research aim	Model Resolution	Result
Zheng et al. (2019)	Effects of total NO ₂ column concentration on socio-economic variables such as permanent population, GDP, secondary industry GDP and its proportion	Provincial and prefectural levels	Provincial unit, R ² : 0.722 to 0.781 Prefectural unit, R ² : 0.497 to 0.643
Shi et al. (2014)	Effects of total nighttime light (TNL) on GDP and EPC	Provincial level	DMSP-OLS: R ² varies from 0.005 to 0.961 NPP-VIIRS: R ² varies from 0.163 to 0.981
Pinkovskiy & Sala-i-Martin (2016)	Effects of Quality-of-Life Indicators, National Accounts, and Survey Means on GDP per capita	National level	R ² varies from 0.02 to 0.44
Ahn et al. (2023)	Measuring economic activity using satellite imagery in data-poor countries	National level	R ² varies from 0.39 to 0.84

5.2 NO₂ Explanatory Model

In this model, the aim is to find out the impact of various factors contributing to NO₂ levels in Addis Ababa and Baghdad. Machine learning models are appropriate choices for this task given the large sample size (in millions), and the possible non-linear relationship among features.

Among machine learning models, tree-based algorithms are selected based on empirical suitability, as they offer a good trade-off between accuracy, interpretability, and computational efficiency. We restricted the modelling shortlist to two proven ensemble-tree families as shown in Table 5.2. This dual-algorithm strategy enables a direct comparison between a variance-reduction approach (Random Forest) and a bias-reduction approach (XGBoost), allowing us to select the best-performing model for each city.

Table 5.2 NO₂ Model Architecture Selection

Algorithm	Architecture	Selection Rationale	Key Advantages
Random Forest	<i>Parallel / bagging</i> → each tree is trained on a bootstrap sample and votes independently.	Fast to train, low variance, and effective at capturing non-linear feature interactions with minimal tuning.	Provides a robust baseline and generates clear SHAP-based explanations for stakeholders.
XGBoost	<i>Sequential / boosting</i> → each new tree learns the residual errors of the current ensemble.	Delivers state-of-the-art accuracy on structured data; built-in regularisation helps prevent overfitting.	Achieves lower RMSE; alignment with RF improves interpretability and model trust.

Note – Deep learning architectures were deliberately excluded at this stage due to time and hardware constraints.

5.2.1 Tree-based Model Fine-tuning

Tree-based models consist of multiple decision trees and are well-suited for capturing non-linear relationships and complex feature interactions. However, their performance is highly sensitive to structural hyperparameters, such as tree depth, learning rate, and the number of estimators. In this project, we focus on fine-tuning these models with the following goals:

1. Improve generalisation
2. Reduce computational cost
3. Maintain interpretability

To achieve these objectives, we employed Randomised Search as the hyperparameter optimisation strategy. Unlike exhaustive methods such as grid search, Randomised Search efficiently explores the hyperparameter space by randomly sampling parameter combinations.

This approach is especially useful when computational resources are limited, as it can yield good results more quickly.

After optimising both the Random Forest and XGBoost models, we identified their optimal hyperparameter configurations. The corresponding performance metrics are presented in Table 5.3 for Addis Ababa. Due to the large dataset size in Baghdad (over four million samples) and limited computational resources, we reduced the complexity of the hyperparameter search. The resulting model performance is summarised in Table 5.4.

Table 5.3 Preliminary NO₂ Models for Addis Ababa

Model Type	If Scale	RMSE	R ²	Best Parameters
Random Forest	Unscaled	1.842*10 ⁻⁵	0.21495	'n_estimators': 200, 'max_depth': 15, 'max_features': 0.5, 'min_samples_leaf': 4
XGBoost	Unscaled	1.858*10 ⁻⁵	0.20120	'subsample': 1.0, 'min_child_weight': 3, 'max_depth': 12, 'eta': 0.2, 'colsample_bytree': 1.0
XGBoost	Scale X & Y	1.840*10 ⁻⁵	0.21652	'subsample': 0.7, 'min_child_weight': 1, 'max_depth': 8, 'eta': 0.01, 'colsample_bytree': 1.0
XGBoost	Only Scale X	1.858*10 ⁻⁵	0.20114	'subsample': 1.0, 'min_child_weight': 3, 'max_depth': 12, 'eta': 0.2, 'colsample_bytree': 0.7

Table 5.4 Preliminary NO₂ Models for Baghdad

Model Type	If Scale	RMSE	R ²	Best Parameters
Random Forest	Unscaled	1.325*10 ⁻⁴	0.09575	n_estimators: 50, max_depth: 10, max_features: 0.5, min_samples_leaf: 500
XGBoost	Unscaled	1.315*10 ⁻⁴	0.10857	subsample: 0.7, min_child_weight: 5, max_depth: 12, eta: 0.01, colsample_bytree: 0.7
XGBoost	Scale X & Y	1.314*10 ⁻⁴	0.11045	subsample: 0.7, min_child_weight: 5, max_depth: 12, eta: 0.01, colsample_bytree: 0.7
XGBoost	Only Scale X	1.315*10 ⁻⁴	0.10845	subsample: 0.7, min_child_weight: 5, max_depth: 12, eta: 0.01, colsample_bytree: 0.7

5.2.2 Shapley Additive Explanations (SHAP)

Due to the black-box nature of many machine learning algorithms, it is often difficult to interpret the importance of the input features. Shapley Additive Explanations, SHAP provides a principled and theoretically sound way to attribute model output to individual input features, improving model interpretability. It is based on the concept of Shapley values from cooperative game theory, which fairly distributes the "payout" (i.e., model prediction) among all the "players" (i.e., features) that contribute to it. One common way to visualise SHAP values is through a violin-like scatter plot, as shown in the results. The interpretation key is as shown in Table 5.5.

Table 5.5 SHAP Plot Interpretation

Plot element	Meaning
Dot	One single datapoint, in this case, one grid-cell on one day.
X-position	How much this feature increases (► right) or decreases (◄ left) the predicted output compared to the average output.
Colour	The raw feature value for that dot (sample): blue = low red/pink = high.
Row order	Features are sorted from most to least influential by their average absolute SHAP value.
Width of the "violin"	Spread of impacts: wide = feature effect varies a lot across space/time; narrow = stable effect.

5.3 Industrial Production Model

Similar to the NO₂ model, an XGBoost model is run for industrial production to find the impact of different variables on the export proxy. Then, considering the limited number of observations in the data frame for the industrial production model, several statistical methods using a regression equation of the log-log form were examined and tested in order to obtain elasticities. This included ordinary least squares (OLS), fixed effects, random effects, system and difference generalised method of moments (GMM). The final model chosen was system GMM (Arellano-Bover/Blundell-Bond). This model uses a first-differences equation and a levels equation with instrumental variables. This improves the model's efficiency and makes it able to overcome persistence. The first differences equation is given by:

$$\Delta y_{it} = \beta \Delta X_{it} + \gamma \Delta IV_{i,t-m} + \Delta v_{it}$$

Where y represents the dependent variable, X represents the vector of independent variables, β represents the vector of coefficients corresponding to each independent variable, IV represents the instrumental variables, and v represents the error component over group i and time t . The levels equation is given by:

$$y_{it} = \beta X_{it} + \gamma IV_{i,t-m} + \mu_i + v_{it}$$

Here, μ is the time-invariant individual-specific effect.

The analysis is done at the sub-city level at a monthly resolution. As commercial area and industrial area are static variables, their conversion to interaction variables was tested using other time-variant variables such as NO₂. Table 5.6 summarises the finalised model inputs. Variables such as population, road length, etc. were tested in the model but ultimately not included as they showed a high variance inflation factor, which could be due to multicollinearity. This is in line with the substantial correlation found between population (or road length) and nightlights in the correlation matrix in Appendix Figure 5. Their time-invariant nature also discouraged their use in the model as such variables introduce bias in GMM models.

Table 5.6 Industrial Production Model Variables

Dependent variable (y)	Export proxy
Independent variables (X)	NO ₂ , nightlights, points of interest Interaction variable: commercial area*NO ₂ , industrial area*NO ₂
Instrument variables (IV)	Lag of export proxy and lag of NO ₂

5.4 Mobility Model

The mobility model uses TCI as a measure of congestion. The model is employed at the district level in Baghdad on a monthly scale. Two main families of specifications were tested to assess the relationship between mobility and NO₂ levels (along with other explanatory variables). These are regression equations but differ by the target variable and functional form as outlined in Table 5.7. The aim is to check whether normalisation of TCI is necessary to obtain an efficient model with reliable results. Road length is used for normalisation as it accounts for infrastructural supply for mobility.

Table 5.7 Mobility Model Specification Types

Target	Model
$\frac{TCI}{road\ length}$	1. OLS (linear form): a parsimonious benchmark on the congestion intensity measure. 2. OLS (log-log form): turns all coefficients into elasticities (requires log-safe ϵ shift). 3. 3rd-degree polynomial: allows each predictor to enter non-linearly (pure powers only).
TCI	Same three variants, but with TCI instead in order to test the necessity of normalising TCI by road length.

5.4.1 Model Performance

As the dataset is large enough, unlike the industrial production model, the dataset was divided into training and test data to check the model performance. This performance was obtained as shown in Table 5.8.

Table 5.8 Performance of Candidate Mobility Models

Target	Model	Parameters	RMSE _{train}	R ² _{train}	RMSE _{test}	R ² _{test}
$\frac{TCI}{road\ length}$	OLS (linear)	12	5.34	0.72	6.14	0.68
	OLS (log-log)	12	5.09	0.17	5.06	−0.01
	Poly-3	364	2.86	0.92	33.27	−8.48
<i>TCI</i>	OLS (linear)	12	8548952	0.75	9846075	0.80
	OLS (log-log)	12	5.82	0.15	5.74	−0.02
	Poly-3	364	2537683	0.98	59692128	−6.47

The RMSE values for TCI are in the order of 10^7 – 10^8 due to its scale as a monthly aggregate across all roads and all days. R² provides a scale-free metric for comparing models' explanatory power. In some of the models, a negative R² is obtained in the test dataset given the formulation of sklearn for this variable; these values occur in the test set when a model performs worse than simply predicting the average of the target variable. This means the residual error from the model exceeds the variance of the observed data. In our case, models like log–log and 3rd-degree polynomial regressions severely overfit the training data and fail to generalise, resulting in negative test R².

5.3.2 Model Finalisation

Plain OLS is the only specification that generalises; both log-log versions under-fit and the 3rd-degree polynomials catastrophically over-fit. Within the OLS pair, modelling raw TCI delivers the best hold-out R² (≈ 0.80) which is within the acceptable range as per Table 5.1. Thus, a linear OLS model using TCI as the dependent variable is chosen for the final analysis.

6. Results & Discussion

6.1 Model Results

6.1.1 NO₂ Explanatory Model

To gain a deeper understanding of the key drivers influencing NO₂ concentration dynamics in the interested region, we performed NO₂ level explanatory analysis using Random Forest (RF) and XGBoost (XGB) models. Model interpretation was conducted via SHAP values.

The target variable is grid-level NO₂ concentration, with values on the order of 10^{-5} . For feature preprocessing, the RF model utilised raw input data, while all features in the XGB model were normalised to the $[0, 1]$ range. This decision was based on comparative performance evaluation across different preprocessing strategies, including unscaled inputs, scaling only the features, and scaling both features and the target variable.

i. Addis Ababa

The SHAP value violin plots of the best two models for Addis Ababa are shown in Figure 6.1 and 6.2. Both models consistently identified the lagged NO₂ concentration in neighboring grids as the most influential predictor, highlighting the significant role of spatial diffusion in pollutant dynamics. Additionally, features reflecting human activity levels, such as cloud cover, population, and nightlights ranked highly in both models, underscoring the non-negligible contribution of anthropogenic factors to air pollution levels.

Of particular interest is the strong influence of nighttime light intensity observed in both models. This variable commonly serves as a proxy for night-time economic activity and population aggregation, capturing composite effects of commercial vibrancy, traffic density, and industrial lighting. SHAP analysis reveals that areas with higher NTL values (the reddish regions in the SHAP plots) are positively associated with elevated NO₂ concentrations, indicating that mobile pollution sources such as night-time traffic and industrial emissions may play a critical role in the spatio-temporal distribution of NO₂.

When it comes to meteorological factors, the land surface temperature stands out as an important variable in the XGB model, but it shows little influence in the RF model. This difference may be due to the effect of feature normalisation in XGB, which can make small changes in temperature appear more impactful. It may also reflect the more complex and nonlinear role temperature plays in NO₂ behaviour. On one hand, higher temperatures can boost air circulation and speed up chemical reactions that involve NO₂. On the other hand, in some situations, high temperatures can increase the formation of ground-level ozone, which may reduce NO₂ levels. This kind of two-sided effect might be better captured by the XGB model, which is more sensitive to subtle patterns in the data.

In addition, road-related variables closely linked to transportation activity (e.g., residential road length, primary road length, total road length) demonstrate medium-to-high importance in both models. Road length not only reflects the density of transportation infrastructure but also indirectly indicates the frequency of vehicular movement and emission sources. Particularly in the RF model, where no normalisation was applied, road-related features with larger value scales

exhibit stronger SHAP responses, suggesting a stable contribution to NO₂ levels.

In summary, both models reveal the multifactorial drivers of NO₂ concentration variability, including spatial lag effects, night-time economic activity, meteorological conditions, and transportation infrastructure. While the exact rankings of feature importance differ slightly between models, the core influential variables remain consistent. These findings offer actionable insights for urban air pollution mitigation, suggesting that policy efforts should focus on controlling emissions in high-NTL areas, regulating night-time economic activities, and fostering regional coordination in response to spatial diffusion of pollutants.

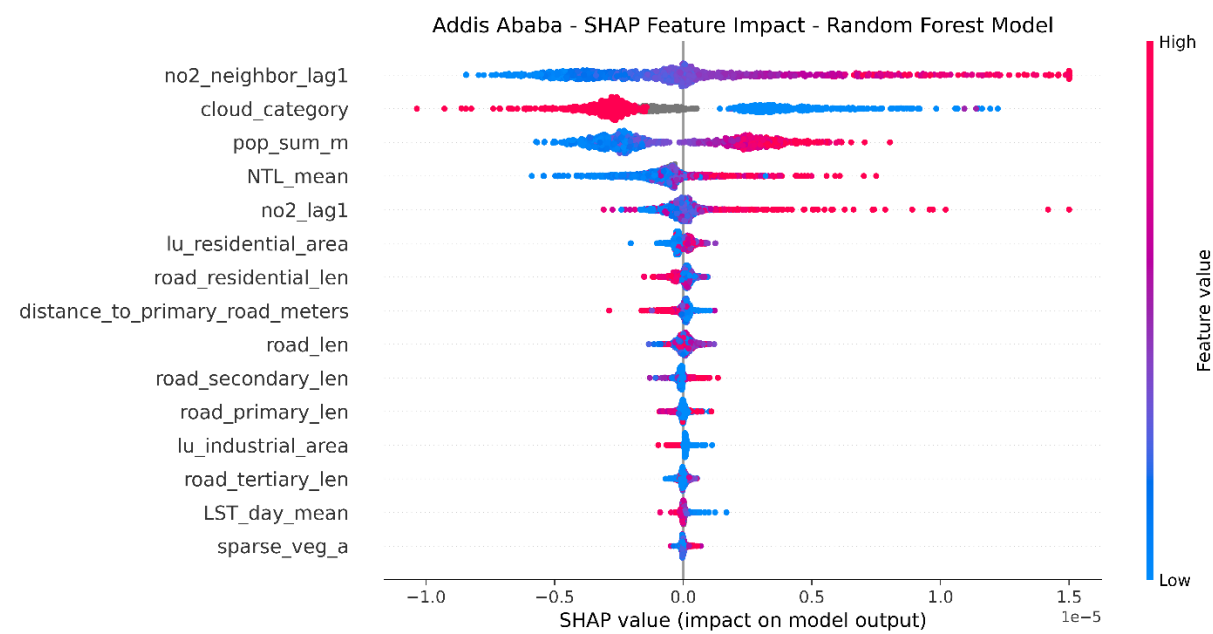


Fig. 6.1 Addis Ababa SHAP Feature Impact - Random Forest Model

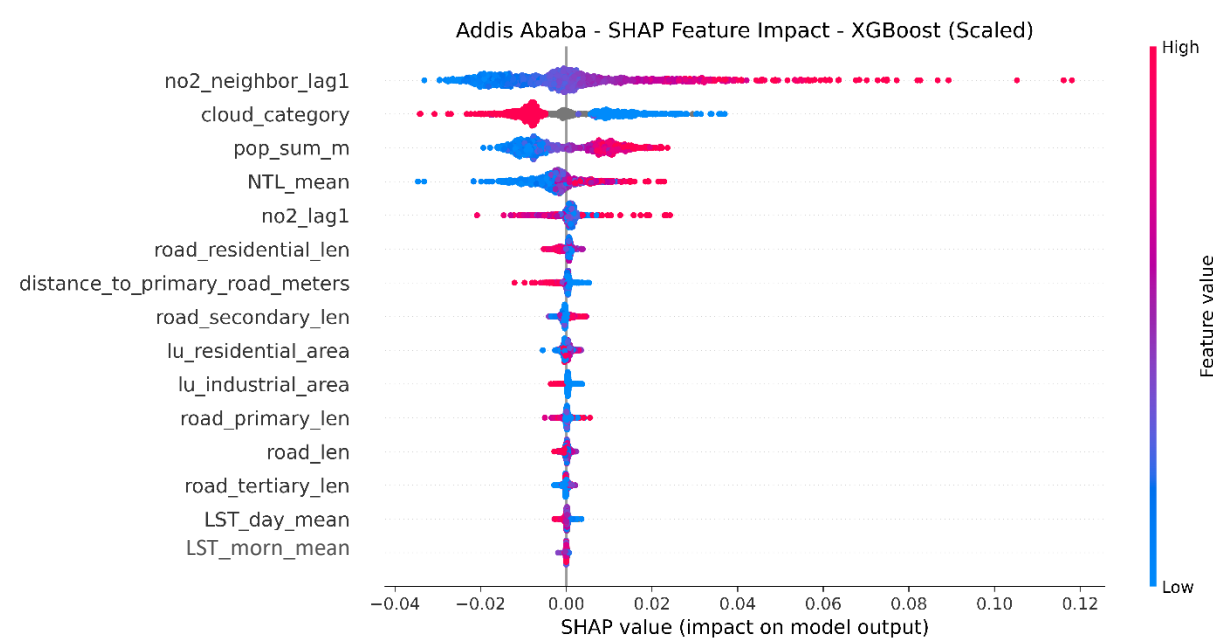


Fig. 6.2 Addis Ababa SHAP Feature Impact - XGBoost (Scaled)

ii. Baghdad

The SHAP plots in Figures 6.3 and 6.4 show that similar to Addis Ababa, both models for Baghdad consistently highlight spatial lag effects, confirming the critical role of regional pollutant spillover. Indicators of human activity, such as nighttime light intensity, also show strong and consistent influence across the models, reflecting the contribution of nocturnal economic and transportation activity to urban NO₂ emissions. These shared findings underscore a common underlying structure in both models, where spatial dependence and anthropogenic factors are central to explaining pollutant variation.

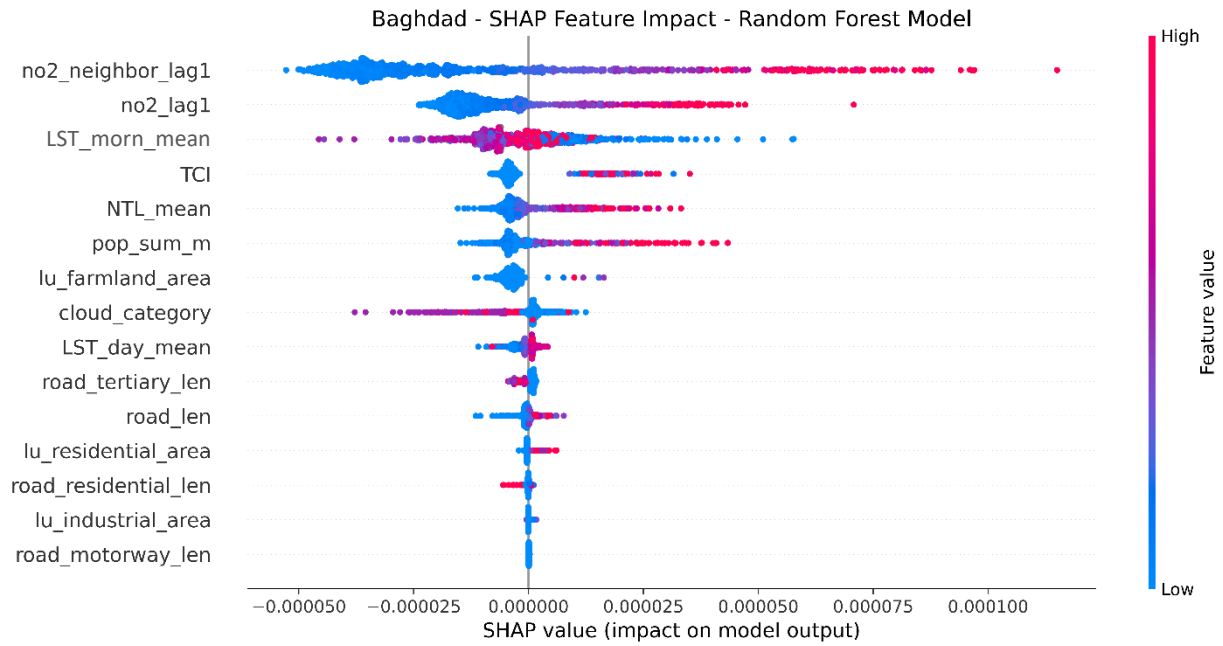


Fig. 6.3 Baghdad SHAP Feature Impact - Random Forest Model

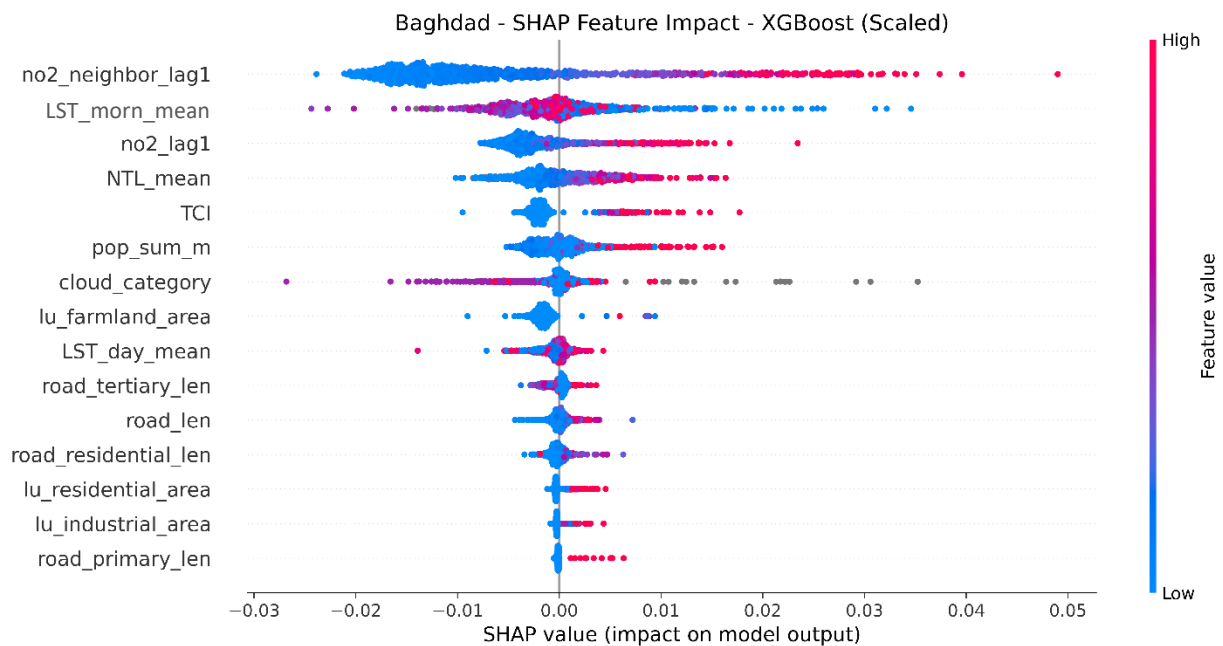


Fig. 6.4 Baghdad SHAP Feature Impact - XGBoost (Scaled)

In the case of Baghdad, the XGBoost model further emphasises the role of dynamic environmental factors. In particular, land surface temperature (10:30 AM) emerges as a highly influential feature, likely due to its role in modulating vertical mixing, atmospheric stability, and the photochemical transformation of NO₂. Elevated temperatures can reduce pollutant dispersion and intensify local accumulation of NO₂, especially under stagnant meteorological conditions. The model also ranks Traffic Congestion Intensity (TCI) among the top predictors, capturing the real-time impact of mobility bottlenecks on transport emissions. These results suggest that the XGBoost model is especially sensitive to temporally variable features, which aligns with its ability to model complex nonlinear interactions when inputs are normalised.

By contrast, the Random Forest model shows a tendency to prioritise structural and infrastructural features, such as residential road length and total road length, over meteorological or dynamic urban variables. This is likely influenced by the use of raw feature scales, which may bias the model toward variables with inherently larger numeric scales. As a result, Random Forest may underrepresent the relative impact of high-frequency or small-scale fluctuations (e.g., TCI or temperature), and instead overemphasise more stable, cumulative spatial attributes. Nevertheless, RF still captures the importance of key variables such as nightlights and lagged NO₂ value for neighbouring cells, suggesting broad alignment in the most essential predictors.

The divergence in feature prioritisation between the two models reveals their complementary strengths: XGBoost appears better suited for capturing short-term, high-variability drivers of NO₂ (e.g., meteorology and traffic dynamics), while Random Forest offers a more stable representation of long-term or structural determinants (e.g., built environment and infrastructure). These differences are not only methodological, influenced by model architecture and preprocessing pipelines, but also conceptual, highlighting how different modeling approaches may uncover distinct but meaningful layers of insight in urban air pollution dynamics.

Overall, the results from both models illustrate a multifactorial landscape driving NO₂ variability in Baghdad, shaped by spatial spillover, human mobility, and atmospheric regulation.

6.1.2 Industrial Production Model

The System GMM model estimates the relationship between exports from sub-cities in Addis Ababa and a range of environmental and urban activity variables.

We first built an XGBoost model using NO₂ along with other economic proxies (e.g., nightlights and land use features), and identified the best-performing hyperparameter configuration. Using this optimised setup, we then re-trained the model excluding NO₂ to assess its marginal contribution. The results are as outlined in Table 6.1. As shown in this table, removing NO₂ leads to a substantial decline in model performance: RMSE increases, while R² drops from 0.468 to 0.213. This clearly highlights NO₂'s role in enhancing model accuracy and supports its value as a complementary proxy for economic activity.

Table 6.1 Testing NO₂ Significance for Industrial Production

Model Type	Including NO ₂	RMSE	R ²	Best Parameters
XGBoost	Yes	3.16*10 ⁸	0.468	'subsample': 1.0, 'min_child_weight': 5, 'max_depth': 6, 'eta': 0.2, 'colsample_bytree': 0.7
XGBoost	No	3.84*10 ⁸	0.213	'subsample': 0.8, 'min_child_weight': 5, 'max_depth': 6, 'eta': 0.1, 'colsample_bytree': 0.7

The SHAP plot obtained through XGBoost is as shown in Figure 6.5. Both NO₂ and nightlights are found to be strong predictors of industrial production with NO₂ ranking as the most important feature. Points of interest have a substantial impact, followed by road length, population, and industrial area. For commercial area, the plot shows that larger areas have negative SHAP values while residential areas have the least effect. Different land use types (industrial, commercial, residential) have relatively small effects, suggesting that single land use types have limited predictive power for economic activity and need to be combined with other features.

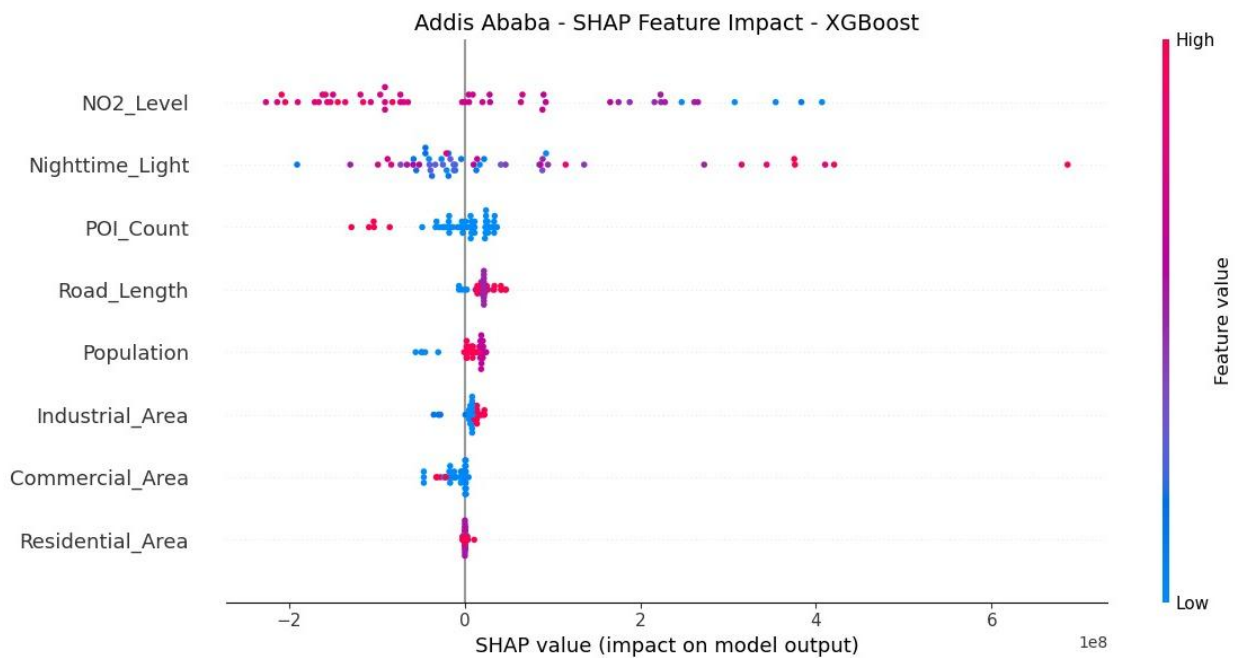


Fig. 6.5 Industrial Production SHAP Feature Impact - XGBoost

The system GMM model was tested by excluding NO₂, but the model failed, indicating inadequateness of data variables. The results of the system GMM industrial production model with NO₂ are as shown in Table 6.2. The results indicate a nuanced relationship. On its own, NO₂ has a positive effect on exports (coefficient = 1.78), suggesting that higher economic activity, which is often associated with greater pollution, may correlate with increased exports. The model suggests that a 1% increase in NO₂ is associated with a 1.78% increase in export values.

However, when NO₂ is combined with commercial and industrial land use, the interaction terms become negative. This implies that in areas with substantial commercial or industrial land, higher NO₂ concentrations are associated with lower exports, potentially due to congestion, reduced productivity, or adverse health impacts affecting labour productivity. It could also indicate that industries may prefer having their operations away from highly urbanised areas. Furthermore, this pattern suggests a threshold or diminishing returns effect, implying a non-linear relationship between NO₂ and industrial production. This is also evidenced by the SHAP plot. At lower levels of air pollution, NO₂ may simply be a byproduct of growing economic activity. However, beyond a certain point, especially in already dense or industrialised areas, the negative externalities of pollution, such as health impacts, reduced worker productivity, or logistical inefficiencies, begin to outweigh the economic benefits.

Table 6.2 Industrial Production Model Results

Variable	Coefficient	Interpretation
NO ₂	1.7849	Positive effect of NO ₂ on exports.
	(1.0615)	
Nightlights	2.9627	Nighttime lights are positively associated with exports.
	(1.1665)	
Points of Interest	6.5591	Denser POIs associated with higher exports.
	(1.3026)	
(Commercial area)*NO ₂	-1.0064	Commercial areas with high NO ₂ negatively affect exports.
	(0.1979)	
(Industrial area)*NO ₂	-2.1184	Industrial areas with high NO ₂ negatively affect exports.
	(0.5061)	
Validation Tests (p-value)		
Sargan test	0.9626	p > 0.05 satisfied (implies instruments are valid)
AB AR (1)	7.287*10 ⁻⁵	p < 0.05 satisfied (implies first order autocorrelation is present)
AB AR (2)	0.0812	p > 0.05 satisfied (implies no second order autocorrelation, instruments are not mis-specified)
Wald test	1.1108*10 ⁻⁵	p < 0.001 satisfied (joint significance of all regressors confirmed)

Note: Numbers in brackets indicate standard error values.

Nighttime lights and points of interest show positive associations with exports, acting as proxies for economic vibrancy and urban density. Accordingly, a 1% increase in nightlights intensity across a sub-city is associated with a 2.96% increase in export values, and this value increases to 6.55% for points of interest.

The model diagnostics confirm its statistical reliability. The Sargan test supports instrument validity ($p = 0.96$), while the absence of second-order autocorrelation ($p = 0.08$) suggests consistent moment conditions. Overall, while modest pollution levels may accompany economic growth, their interaction with dense land use can counteract potential gains, highlighting a trade-off between urban expansion and environmental burden.

In conclusion, NO₂ can be a good measure of industrial production in a region. The findings suggest that NO₂ can be effectively used not only as an input feature within predictive models but also as a standalone spatial proxy for economic activity intensity in similar urban environments. However, the relationship between the NO₂ and industrial production may not necessarily always be linear, other variables and the specific context of the region must be considered as well.

6.1.3 Mobility Model

The results obtained from the OLS model for the top contributors to TCI are as outlined in Table 6.3. Both NO₂ and TCI are aggregated spatial-temporal quantities. TCI aggregates congestion minutes for every road cell across all days of the month. The NO₂ levels variable averages daily values for each grid cell, then sums across all cells in a district. Thus, the coefficient captures total congestion response to total pollution, naturally leading to a large absolute number as shown in the results table.

To make this interpretable in relative terms, we compute a point elasticity:

$$\varepsilon_{NO_2, TCI} = \beta_{NO} \frac{\bar{x}}{\bar{y}} \approx 0.25$$

For a typical Baghdad district and month, a 1% increase in average NO₂ concentration is associated with roughly a 0.25% increase in total monthly congestion (TCI). Because the regression is in levels, this 0.25 figure is a local effect, it holds near current mean conditions. Elasticity would vary in areas with very different baseline pollution or congestion levels.

NO₂ is a by-product of combustion (traffic, power generation, industrial boilers). Its strong positive coefficient indicates that air-quality deterioration is tightly linked to higher mobility demand and hence economic throughput in Baghdad. Simpler proxies like lags or neighbouring NO₂ add little once the mean level is in the regression; they were therefore removed to avoid collinearity.

In summary, a transparent, district-month OLS confirms that NO₂ is a reliable real-time pulse of economic activity, explaining ~80 % of congestion variance while remaining fully interpretable for policy use.

Table 6.3 Mobility Model Results

Rank	Predictor	Coefficient	Interpretation
1	NO ₂	+53,986,362	A 1 unit rise in monthly average NO ₂ is associated with a +53.9 million change in aggregated TCI for that sub-city in that month. The large magnitude arises because both NO ₂ and TCI are aggregated sums (TCI counts all congested travel-time seconds across all roads for an entire month).
2	Points of Interest	+930,026	One additional point-of-interest correlates with ~0.93 million extra TCI units (more activity → more congestion).
3	Land surface temperature (day)	−462,200	Hotter daytime surface temperatures slightly reduce monthly congestion—potentially fewer trips in extreme heat.
...	Road lengths (primary, motorway, trunk, etc.)	−206,000 → −50,000	More road supply in each category correlates with lower TCI, as expected.
...	Land-use areas (grassland, cropland, built-up, ...)	order 10 ² → 10 ^{−1}	Marginal effects are small once major drivers are accounted for.

6.2 Limitations and Future Directions

This study offers promising insights into how satellite-based indicators, particularly NO₂, can help estimate economic activity in cities where conventional data is limited. While all the tested models are validated to ensure reliability and robustness, there are some limitations that should be acknowledged to guide future work. These can be summarised as follows:

1. Data availability and resolution remain major hurdles. While global satellite datasets are designed to cover all regions, cities like Baghdad and Addis Ababa still suffer from patchy observations. Factors such as persistent cloud cover and retrieval issues led to missing values in variables like NO₂, which had to be estimated through spatial and temporal interpolation. While these methods were applied carefully, they introduce a layer of uncertainty that could affect model outcomes. This highlights a broader challenge; the need for more consistent and high-quality satellite data in underrepresented regions.

2. The analysis covers a relatively short time period of just under two years (2023–2024). This was sufficient for model training and testing, but it limits the ability to detect longer-term patterns or seasonal effects. Expanding the analysis to at least 3 to 5 years of continuous data would help improve model stability and provide a more comprehensive view of economic cycles and environmental dynamics, although it may increase computational demand.
3. Ground-truth economic data is often missing at the sub-city level, especially in lower-income contexts. This study used practical proxies, exports for Addis Ababa and traffic congestion intensity (TCI) for Baghdad, but more accurate calibration would require matching satellite indicators with standard economic measures like GDP, industrial output, or electricity consumption where those are available. Furthermore, the limited nature of the exports data leads to fewer time steps for running a dynamic panel model. While system GMM is built to withstand this, it does interfere with the reproducibility of the results due to a lack of assurance on whether heterogeneity is captured satisfactorily by the data. The use of city level variables may resolve this issue, subject to data availability.
4. While system GMM is a good fit for our industrial production data, the model using R (or Stata) is not directly reproducible in Python. This may be an issue for researchers looking for reproducibility using different software. This can be resolved through the use of Machine Learning models such as XGBoost, as attempted in this project.

A practical recommendation going forward is to train models in regions where both the explanatory variables (e.g. NO₂, NTL, land use) and the economic indicators are reliable, and then apply these models in similar settings where only the satellite features are available. This transfer approach allows for broader applicability while maintaining rigour. To ensure that such models generalise well, they should be tested against alternative benchmarks, such as mobility trends or TCI, grid usage, or trade flows, in the target region. At the same time, producing outputs such as elasticities or SHAP-based feature importance can help keep the models interpretable and policy-relevant.

6.3 Conclusion

The extensive literature review, exploratory data analysis, and models all point to the applicability and importance of NO₂ in measuring economic activity resulting from industrial production and mobility. Across two very different case studies, industrial performance in Addis Ababa and urban mobility in Baghdad, NO₂ consistently emerged as one of the most informative features.

Chapter 2 addresses the production and measurement of NO₂ and factors influencing it, as evidenced by the literature. These findings are validated in Chapter 4 through the exploratory data analysis as relationships are drawn and correlations are estimated. The testing of the modelling methods with different combinations of data variables in Chapter 3 and 5 help build the relationship between NO₂ and economic activity. The results of the models in Chapter 6 confirm our hypothesis; the variability in congestion and industrial production can be reliably predicted using NO₂, and in some cases, NO₂ may even be better than traditionally used predictors such as nightlights. Using a combination of predictors to support each other can produce even better results, especially since contextual variability can introduce unexpected deviations.

Machine learning models delivered strong results where detailed spatial data was available, particularly in the NO₂ explanatory model. However, when data was more aggregated, as in the economic models, simpler methods like ordinary least squares (OLS) performed nearly as well and offered a clearer view of how different variables influenced the outcome. In the Addis Ababa model, NO₂ had an estimated elasticity of +1.78, meaning a 1% increase in NO₂ levels was associated with a 1.78% increase in monthly exports. In Baghdad, the NO₂ elasticity with respect to traffic congestion intensity (TCI) was +0.25%, reinforcing its relevance even in settings where economic activity is more mobility-driven. This suggests that for many real-world applications, especially those involving public or institutional stakeholders, interpretable models remain a critical asset.

Given its advantages of global coverage, frequent updates, and low acquisition costs, satellite-derived NO₂ data holds particular promise for application in low- and middle-income countries, especially in urban areas where economic data may be sparse, outdated, or unreliable. This includes regions characterised by informal economies or affected by conflict, where conventional economic statistics are often insufficient. Thus, NO₂ offers a valuable and practical tool for monitoring and analysing economic activity in data-scarce contexts.

Appendix

Fig. 1 Population per Cell: Boxplot

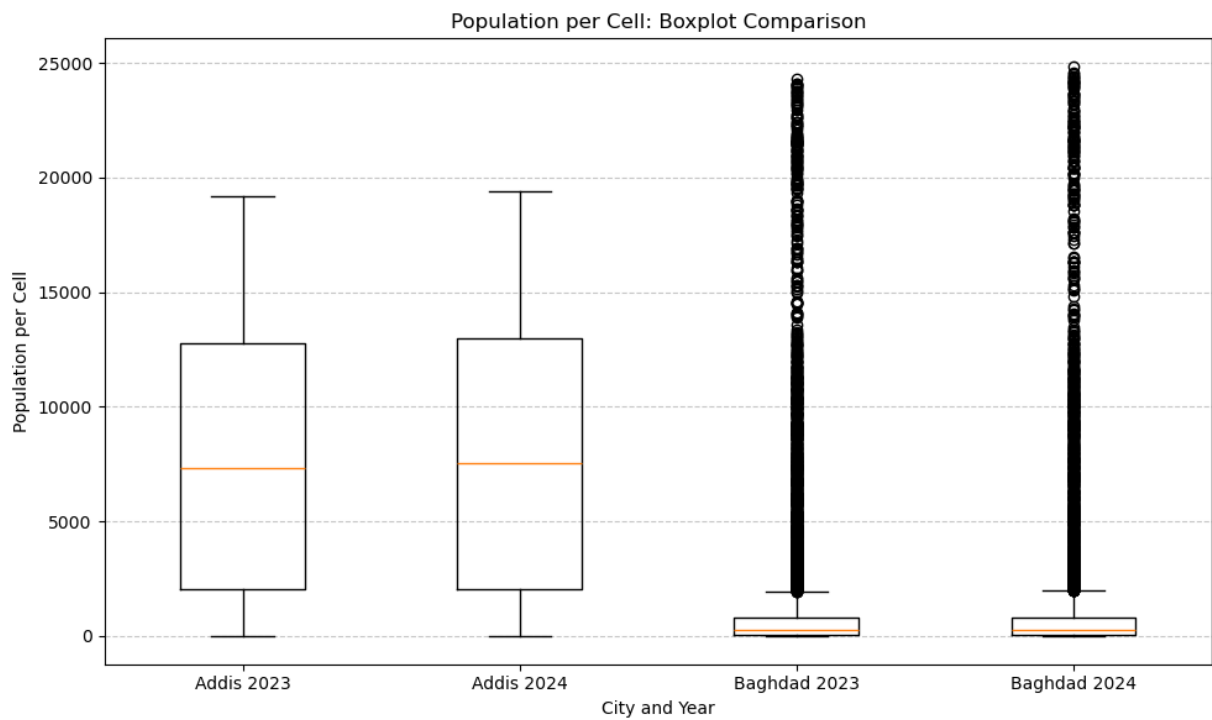


Fig. 2 Population per Cell: Histogram

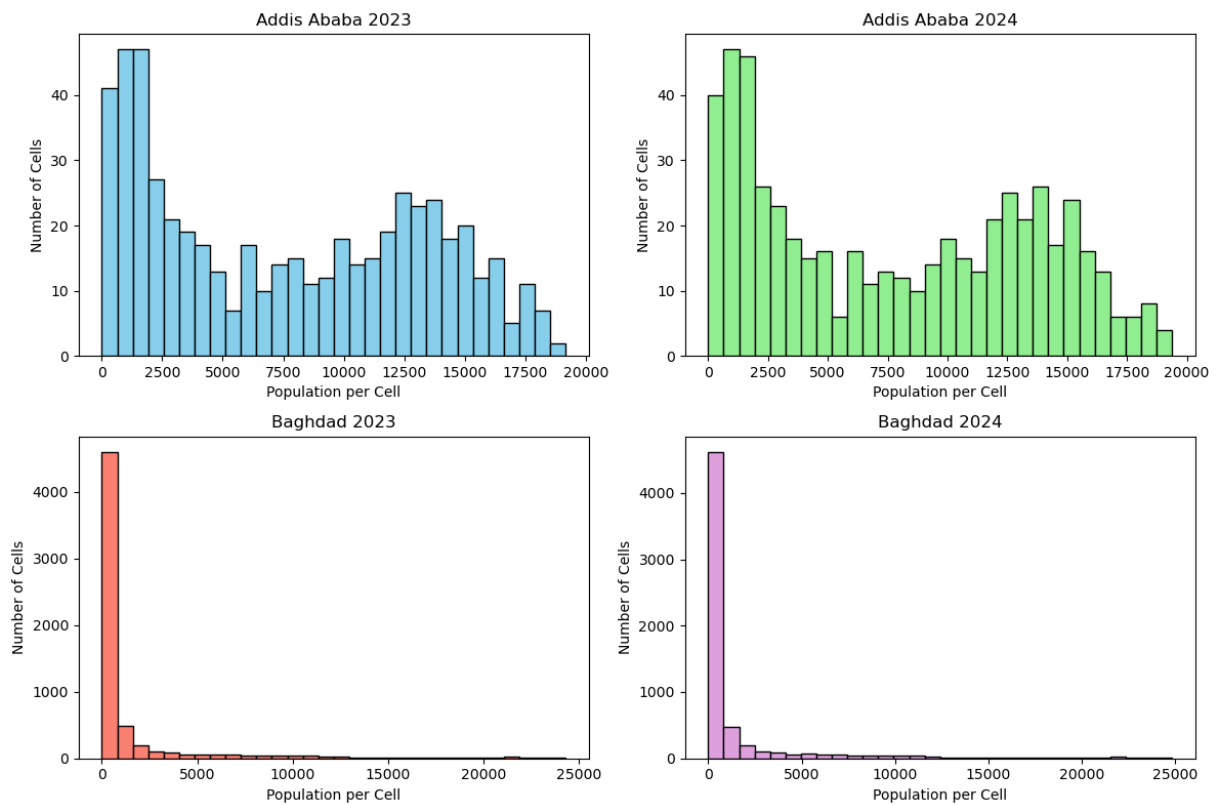


Fig. 3 Addis Ababa Subregional Daily NO₂ Distribution (2023)



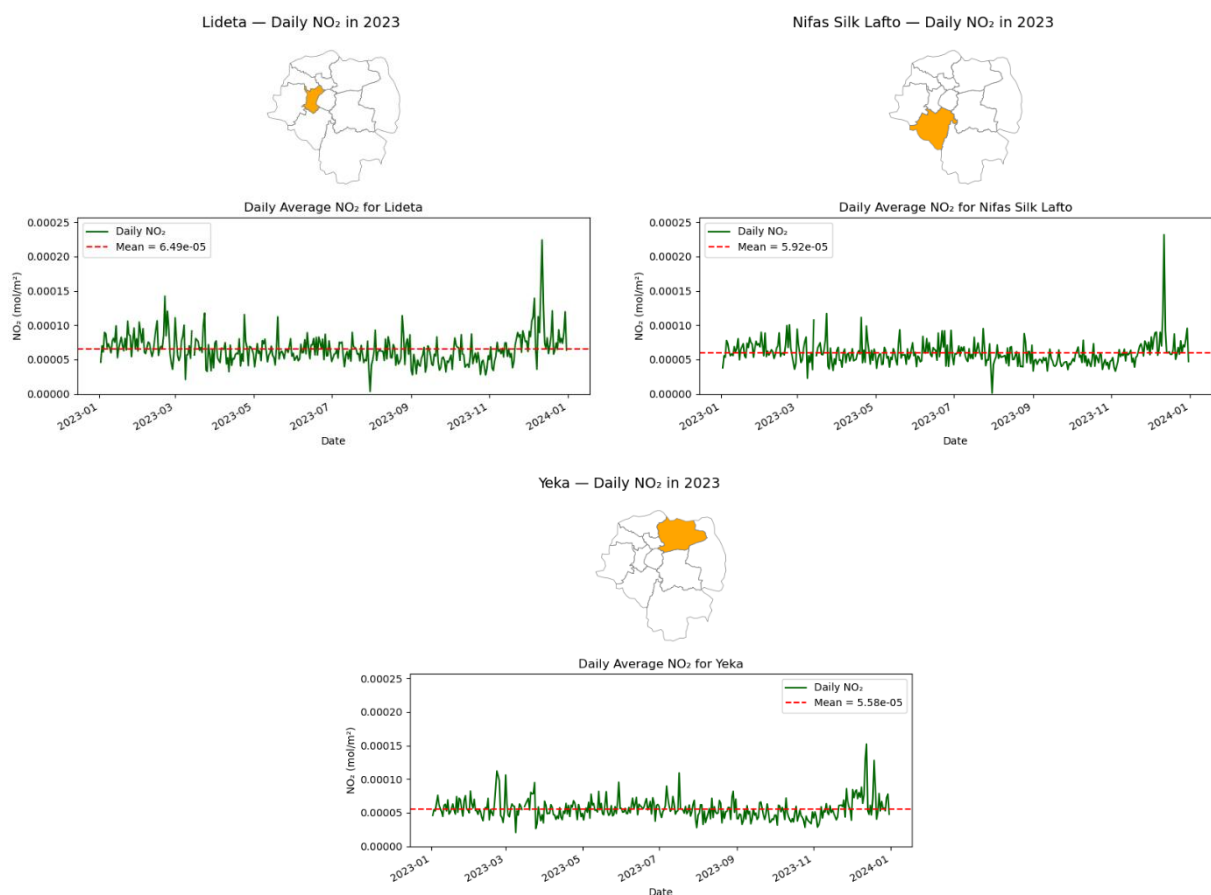
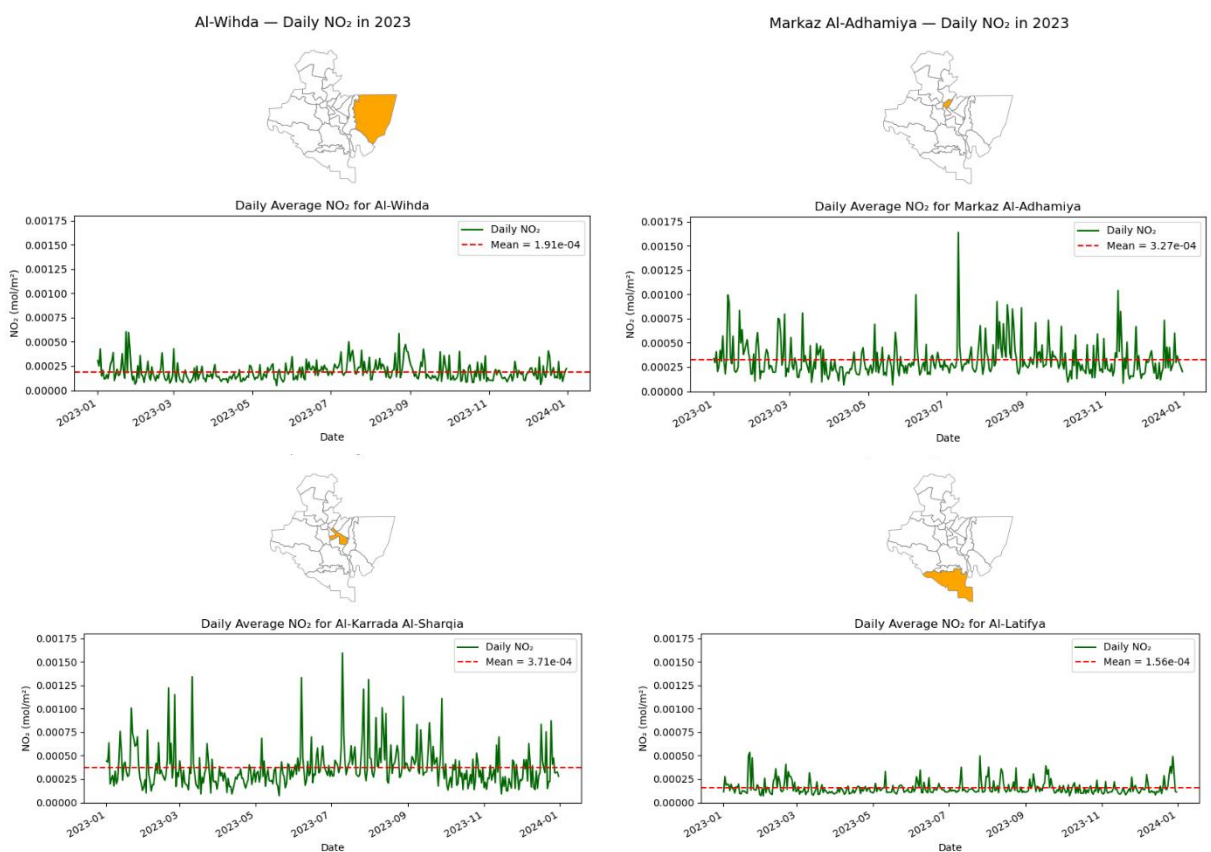
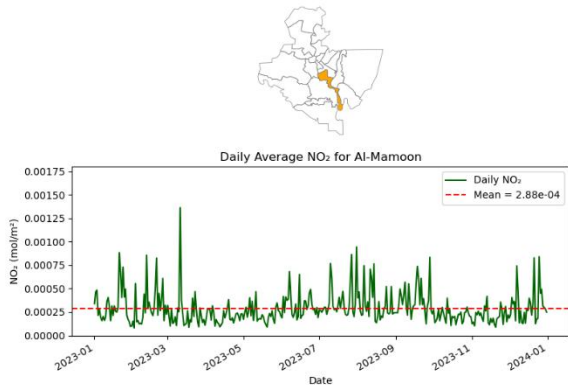


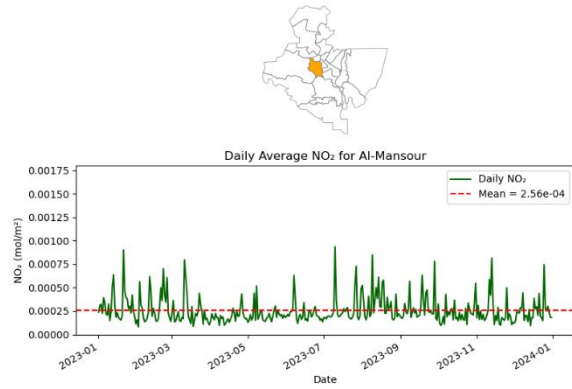
Fig. 4 Baghdad Subregional Daily NO₂ Distribution (2023)



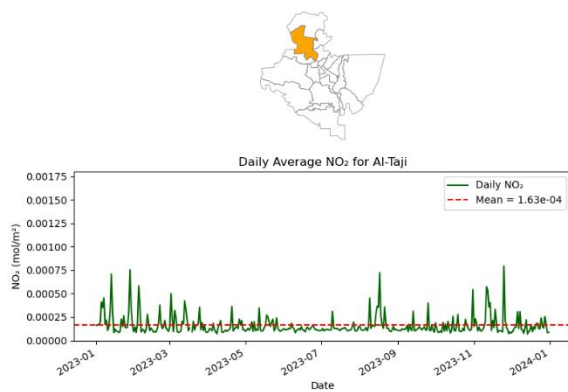
Al-Mamoon — Daily NO₂ in 2023



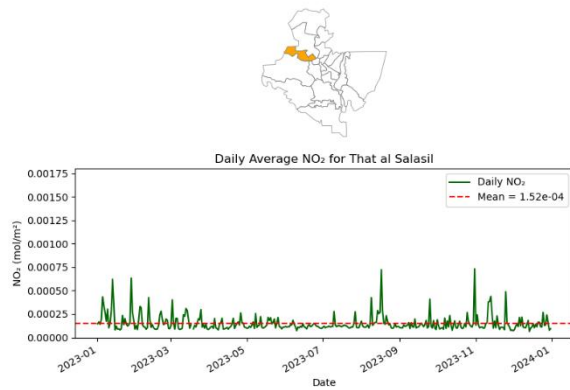
Al-Mansour — Daily NO₂ in 2023



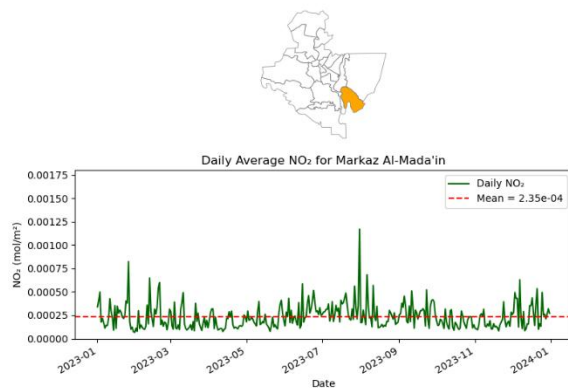
Al-Taji — Daily NO₂ in 2023



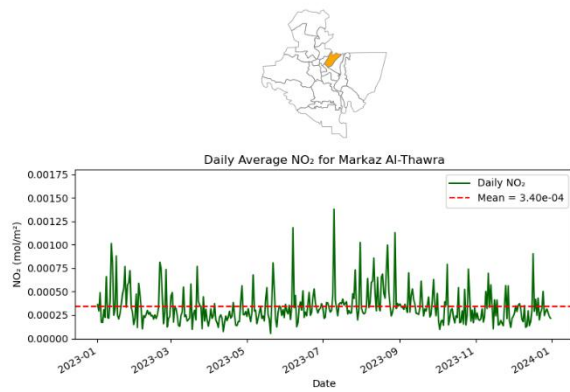
That al Salasil — Daily NO₂ in 2023



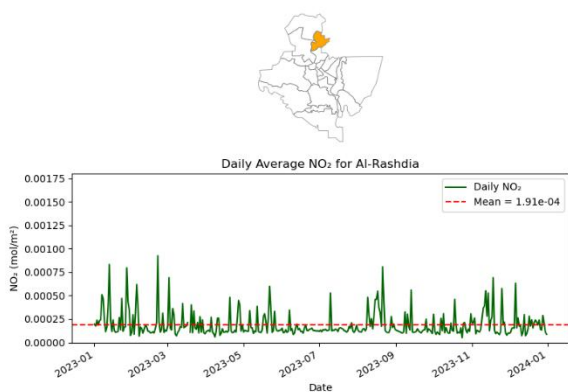
Markaz Al-Mada'in — Daily NO₂ in 2023



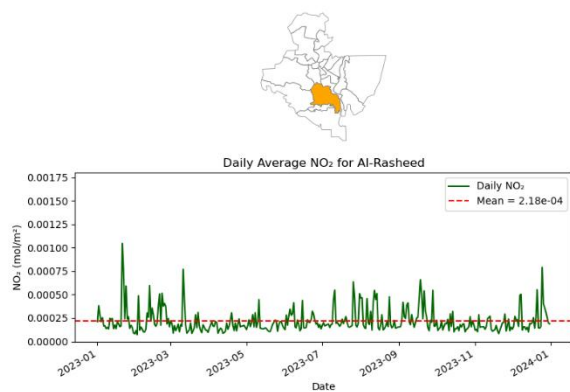
Markaz Al-Thawra — Daily NO₂ in 2023



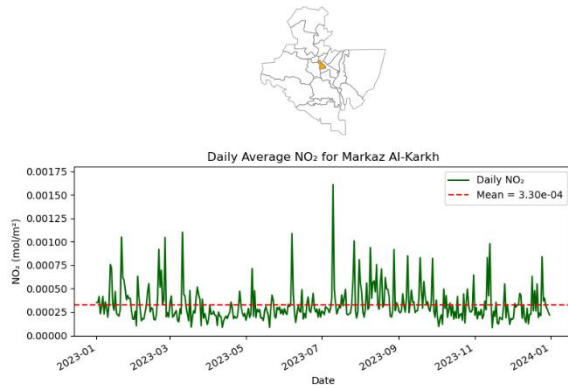
Al-Rashdia — Daily NO₂ in 2023



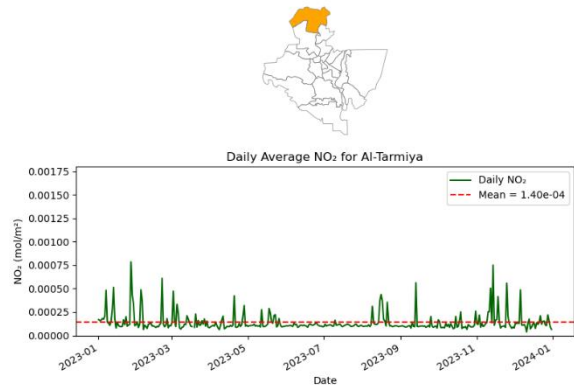
Al-Rasheed — Daily NO₂ in 2023



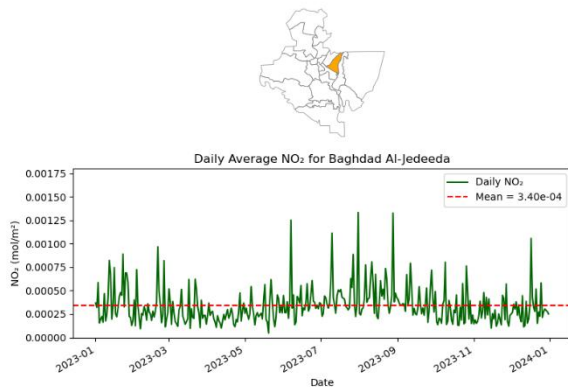
Markaz Al-Karkh — Daily NO₂ in 2023



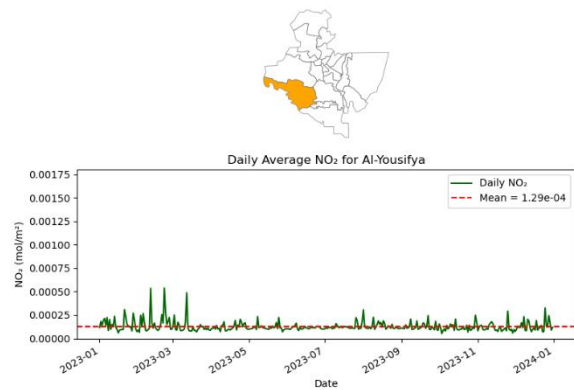
Al-Tarmiya — Daily NO₂ in 2023



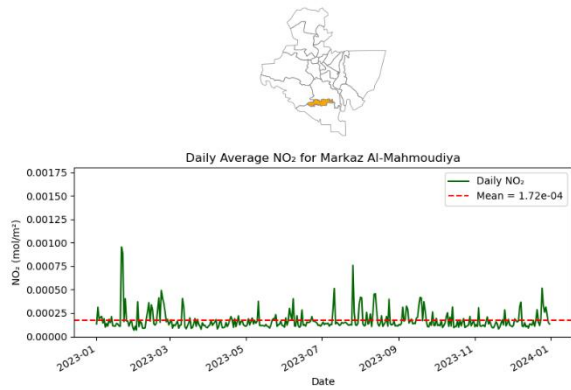
Baghdad Al-Jedeeda — Daily NO₂ in 2023



Al-Yousifya — Daily NO₂ in 2023



Markaz Al-Mahmoudiya — Daily NO₂ in 2023



Markaz Al-Kadhimiya — Daily NO₂ in 2023

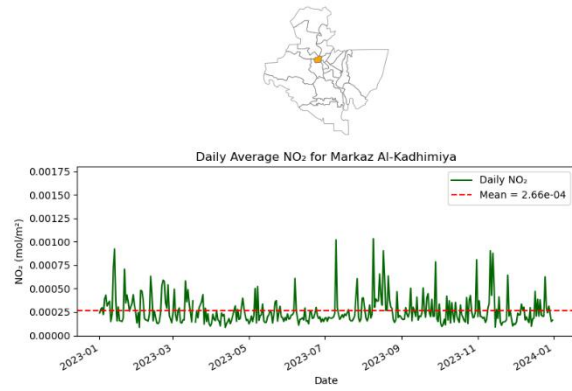


Fig. 5 Addis Ababa Correlation Matrix

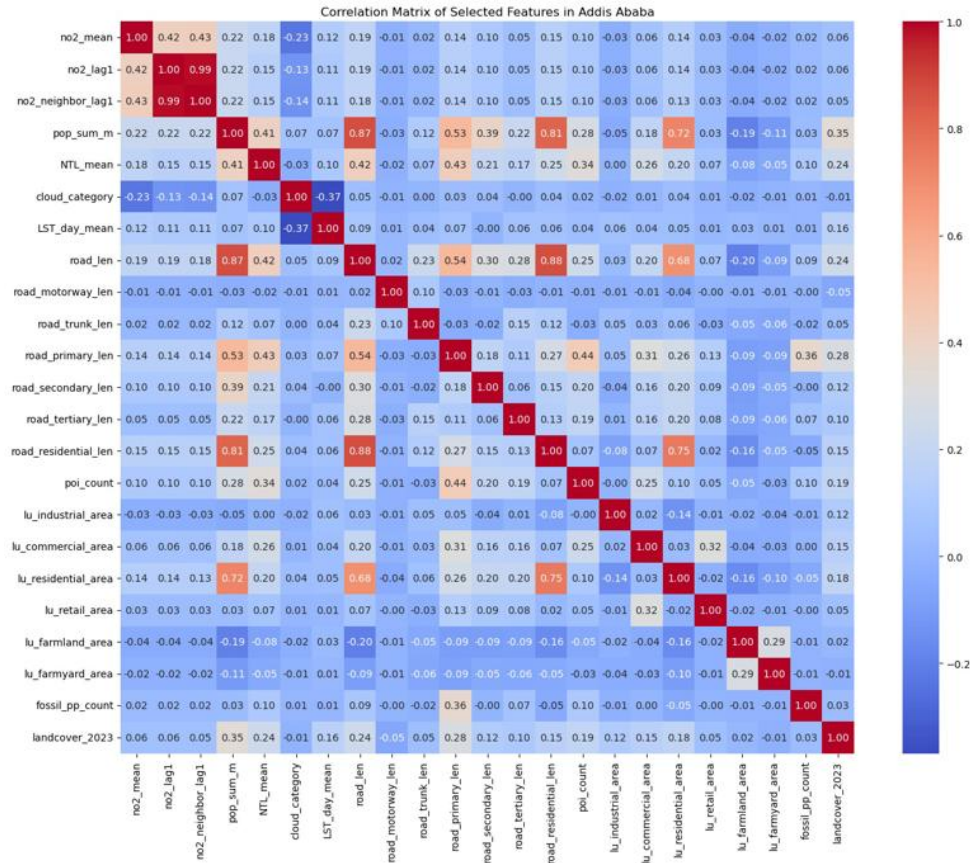
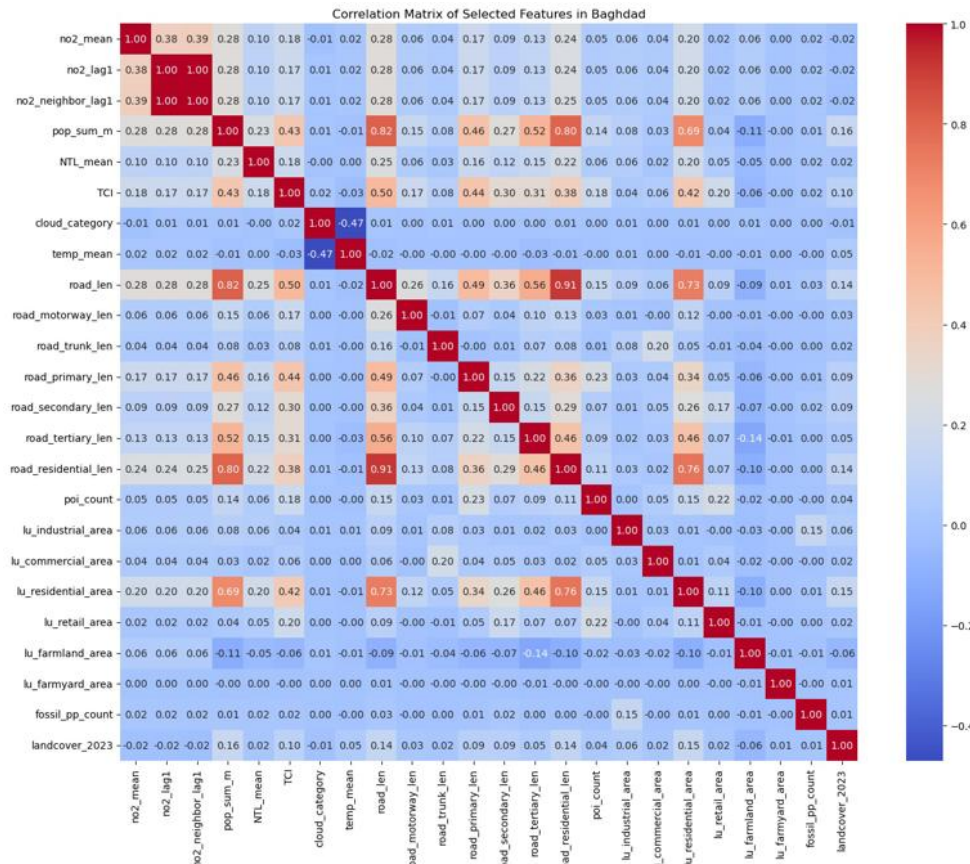


Fig. 6 Baghdad Correlation Matrix



References

- Achyunda, R., Putra, A. and Arini, S. (n.d.). Measuring the Economics of a Pandemic: How People Mobility depict Economics? An Evidence of People's Mobility Data towards Economic Activities. [online] Available at: <https://www.imf.org/-/media/Files/Conferences/2020/8th-stats-forum/paper-rendra-putra-and-silvia-arini.ashx> [Accessed 27 May 2025].
- Ahn, D., Yang, J., Cha, M., Yang, H., Kim, J., Park, S., Han, S., Lee, E., Lee, S. and Park, S. (2023). A human-machine collaborative approach measures economic development using satellite imagery. *Nature Communications*, [online] 14(1), p.6811. doi:<https://doi.org/10.1038/s41467-023-42122-8>.
- Akinlo, A.E. (2008). Energy consumption and economic growth: Evidence from 11 Sub-Saharan African countries. *Energy Economics*, 30(5), pp.2391–2400. doi:<https://doi.org/10.1016/j.eneco.2008.01.008>.
- Alder, S., Croke, K., Duhaut, A., Marty, R. and Vaisey, A. (2023). The Impact of Ethiopia's Road Investment Program on Economic Development and Land Use Evidence from Satellite Data. [online]. Available at: <https://documents1.worldbank.org/curated/en/099332404062230683/pdf/IDU073a7158605532046490b712098aed9008539.pdf>.
- Al-Hameedi, W.M.M., Chen, J., Faichia, C., Al-Shaibah, B., Nath, B., Kafy, A.-A., Hu, G. and Al-Aizari, A. (2021). Remote Sensing-Based Urban Sprawl Modeling Using Multilayer Perceptron Neural Network Markov Chain in Baghdad, Iraq. *Remote Sensing*, 13(20), p.4034. doi:<https://doi.org/10.3390/rs13204034>.
- Alshammari, L., Ridha, S., & Al-shammari, M. (2025). Monitoring the urban heat island effect in Baghdad using Sentinel-3 and OSM data integration for sustainable urban planning. *Journal of Applied Engineering Science*, 23, 151–164. 10.5937/jaes0-54589
- Bundervoet, T., Maiyo, L. and Sanghi, A. (2015). Policy Research Working Paper 7461 Bright Lights, Big Cities Measuring National and Subnational Economic Growth in Africa from Outer Space, with an Application to Kenya and Rwanda. [online] Available at: <https://documents.worldbank.org/curated/en/912151468188369841/pdf/WPS7461.pdf> [Accessed 27 May 2025].
- Chaichan, M.T., Kazem, H.A. and Abed, T.A. (2016). Traffic and outdoor air pollution levels near highways in Baghdad, Iraq. *Environment, Development and Sustainability*, 20(2), pp.589–603. doi:<https://doi.org/10.1007/s10668-016-9900-x>.
- Chakraborty, S., Stokes, E.C. and Alexander, O., 2020. Global urban activity changes from COVID-19 physical distancing restrictions. *Scientific Data*, 12(1). <https://doi.org/10.1038/s41597-025-04398-x>.
- Chen, C., He, X., Liu, Z., Sun, W., Dong, H. and Chu, Y. (2020). Analysis of regional economic development based on land use and land cover change information derived from Landsat

imagery. *Scientific Reports*, [online] 10, p.12721. doi:<https://doi.org/10.1038/s41598-020-69716-2>.

Emma Wadie Hobson. (2018). Ethiopia - Enhancing Economic Development and Job Creation in Addis Ababa - the role of the city administration : Enhancing Economic Development and Job Creation in Addis Ababa - The Role of the City Administration. World Bank. [online] Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/828941529524223693/enhancing-economic-development-and-job-creation-in-addis-ababa-the-role-of-the-city-administration> [Accessed 9 Jun. 2025].

European Institute on Economics and the Environment (2021) Effectiveness, spillovers, and well-being effects of driving. Available at: https://media.rff.org/documents/EIEE_WP_21-13.pdf (Accessed: 22 June 2025).

Ezran, I., Morris, S. D., Rama, M., Riera-Crichton, D. (2023). Measuring Global Economic Activity Using Air Pollution. World Bank Group [online] doi:<https://doi.org/10.1596/1813-9450-10445>.

Feyissa, G. et al. (2018) GIS based quantification and mapping of climate change vulnerability hotspots in Addis Ababa - geoenvironmental disasters, SpringerOpen. Available at: <https://geoenvironmental-disasters.springeropen.com/articles/10.1186/s40677-018-0106-4> (Accessed: 22 June 2025).

Follette-Cook, M., Prados, A. and Gupta, P. (n.d.). National Aeronautics and Space Administration Measuring Nitrogen Dioxide from Space. [online] Available at: https://appliedsciences.nasa.gov/sites/default/files/2020-11/COVID_NO2_Final.pdf.

Geddes, J.A., Murphy, J.G., O'Brien, J.M. and Celarier, E.A. (2012). Biases in long-term NO₂ averages inferred from satellite observations due to cloud selection criteria. *Remote Sensing of Environment*, [online] 124, pp.210–216. doi:<https://doi.org/10.1016/j.rse.2012.05.008>.

Ghosh, T., Powell, R., Elvidge, C., Baugh, K., Sutton, P. and Anderson, S. (2010). Shedding Light on the Global Distribution of Economic Activity. *The Open Geography Journal*, [online] 3, pp.147–160. Available at: <https://benthamopenarchives.com/contents/pdf/TOGEOGJ/TOGEOGJ-3-147.pdf>.

Gu, Y. (2022) WAZE Data Reporting, National Transportation Library | Bureau of Transportation Statistics. Available at: <https://ntl.bts.gov/ntl> (Accessed: 26 May 2025).

Hasan, I., Jackowicz, K., Kowalewski, O. and Kozłowski, Ł. (2018). The economic impact of changes in local bank presence. *Regional Studies*, 53(5), pp.644–656. doi:<https://doi.org/10.1080/00343404.2018.1475729>.

Hassoon, A.F. and Roomi, T.O. (2023). EFFECT OF THE CONVECTIVE BOUNDARY LAYER ON RADIOSONDE FLIGHT PATH OVER BAGHDAD AIRPORT STATION (CASE STUDY). *The Iraqi Journal of Agricultural Sciences*, [online] 54(6), pp.1603–1621. doi:<https://doi.org/10.36103/ijas.v54i6.1861>.

Henderson, J.V., Storeygard, A. and Weil, D.N. (2012). Measuring Economic Growth from Outer Space. *American Economic Review*, 102(2), pp.994–1028. doi:<https://doi.org/10.1257/aer.102.2.994>.

iData. (2025). 'Air Pollution in Baghdad: Alarming Data Threatening Our Future'. December, 2024. <https://www.iraqidata.com/en/blog/pollution-article> (Accessed: 22 June 2025).

Iraq Economists Network. (2020). Iraqi Industrial Overview. Available at: <https://iraqieconomists.net/en/wp-content/uploads/sites/3/2020/10/Industrial-Sector-in-Iraq-KAPITA.pdf> [Accessed 9 Jun. 2025].

Ji, X., Yao, Y. and Long, X., 2018. What causes PM2.5 pollution? Cross-economy empirical analysis from socioeconomic perspective. *Energy Policy*, 119, pp.458–472. <https://doi.org/10.1016/j.enpol.2018.04.040>

Lehnert, P., Niederberger, M., Uschi Backes-Gellner and Bettinger, E. (2023). Proxying economic activity with daytime satellite imagery: Filling data gaps across time and space. [online] 2(4). doi:<https://doi.org/10.1093/pnasnexus/pgad099>.

McCarthy, J. J. (2001). Climate change 2001: impacts, adaptation, and vulnerability: contribution of Working Group II to the third assessment report of the Intergovernmental Panel on Climate Change. Cambridge University Press.

Measurement methods and UK monitoring networks for NO₂ Key points. (n.d.). Available at: <https://uk-air.defra.gov.uk/library/assets/documents/reports/aqeg/chapter4.pdf>.

Mossie, W.A. (2022). Understanding financial inclusion in Ethiopia. *Cogent Economics & Finance*, 10(1). doi:<https://doi.org/10.1080/23322039.2022.2071385>.

Pinkovskiy, M. and Sala-i-Martin, X. (2016). Lights, Camera ... Income! Illuminating the National Accounts-Household Surveys Debate. *The Quarterly Journal of Economics*, 131(2), pp.579–631. doi:<https://doi.org/10.1093/qje/qjw003>.

Saadawi, A., Rami al-Salihi, Hiwa Hawrami, Hasan, M.M., Peshraw Mirany, Haghighi, F., Shareef, B., Rejna Alaaldin, Akram, G., Philip, E., Mustafa, H.A., al-Qaisi, M., Ahmad, A., Bachar, A., Davis, P., Mustafa al-Saray, Wissam Massify, Wali, Z.Z., Chenar Chalak and Soran Naqishbandy (2025). The New Region. [online] The New Region. Available at: <https://thenewregion.com/posts/1576/baghdad-s-traffic-crisis-billions-lost-annually-as-congestion-worsens> [Accessed 10 Jun. 2025].

Sentinel Online. (2023). Sentinel-5 Precursor Level 2 Nitrogen Dioxide - Sentinel Online. [online] Available at: https://sentinels.copernicus.eu/data-products/-/asset_publisher/fp37fc19FN8F/content/sentinel-5-precursor-level-2-nitrogen-dioxide [Accessed 26 May 2025].

Shi, K., Yu, B., Huang, Y., Hu, Y., Yin, B., Chen, Z., Chen, L. and Wu, J. (2014). Evaluating the Ability of NPP-VIIRS Nighttime Light Data to Estimate the Gross Domestic Product and

the Electric Power Consumption of China at Multiple Scales: A Comparison with DMSP-OLS Data. *Remote Sensing*, 6(2), pp.1705–1724. doi:<https://doi.org/10.3390/rs6021705>.

Tabor, K. (2019). THE REGIONAL CONTEXT: FINFINE (ADDIS ABABA) REGIONAL METROPOLITAN SYSTEM 3.1. Introduction. [online] Available at: https://www.researchgate.net/publication/331045460_THE_REGIONAL_CONTEXT_FINFINE_ADDIS_ABABA_REGIONAL_METROPOLITAN_SYSTEM_31_Introduction.

Teklemariam, E.A. and Shen, Z. (2020). Determining transit nodes for potential transit-oriented development: Along the LRT corridor in Addis Ababa, Ethiopia. *Frontiers of Architectural Research*. doi:<https://doi.org/10.1016/j.foar.2020.03.005>.

Tijl Verhoelst, Compennolle, S., Pinardi, G., Lambert, J.-C., Henk Eskes, Eichmann, K.-U., Ann Mari Fjæraa, Granville, J., Sander Niemeijer, Cede, A., Tiefengraber, M., Hendrick, F., Pazmino, A., Bais, A.F., Bazureau, A., Boersma, F., Bognar, K., Dehn, A., Donner, S. and Elokhov, A.S. (2021). Ground-based validation of the Copernicus Sentinel-5P TROPOMI NO₂ measurements with the NDACC ZSL-DOAS, MAX-DOAS and Pandonia global networks. 14(1), pp.481–510. doi:<https://doi.org/10.5194/amt-14-481-2021>.

U Habitat (2025) Urbanization in Ethiopia: Building inclusive and Sustainable Cities, Urbanization in Ethiopia: Building inclusive & sustainable cities. Available at: <https://unhabitat.org/ethiopia> (Accessed: 22 June 2025).

UNDP. (n.d.). Economic Diversification. [online] Available at: <https://www.undp.org/iraq/economic-diversification>.

Van Geffen, J.H.G.M., Eskes, H.J., Boersma, K.F., Maasakkers, J.D. and Veeffkind, J.P. (2019). TROPOMI ATBD of the total and tropospheric NO₂ data products. Report S5P-KNMI-L2-0005-RP, KNMI, De Bilt, The Netherlands. Available at: <https://sentinel.esa.int/documents/247904/2476257/Sentinel-5P-TROPOMI-ATBD-NO2-data-products.pdf>.

Virág, D., Wiedenhofer, D., Baumgart, A., Matej, S., Krausmann, F., Min, J., Rao, N.D. and Haberl, H. (2022). How much infrastructure is required to support decent mobility for all? An exploratory assessment. *Ecological Economics*, 200, p. 107511. doi:<https://doi.org/10.1016/j.ecolecon.2022.107511>.

Wang, G., Peng, W., Zhang, L., Zhang, J. and Xiang, J. (2023). Vegetation EVI changes and response to natural factors and human activities based on geographically and temporally weighted regression. *Global Ecology and Conservation*, 45, pp.e02531–e02531. doi:<https://doi.org/10.1016/j.gecco.2023.e02531>.

Wang, T. and Sun, F. (2022). Global gridded GDP data set consistent with the shared socioeconomic pathways. *Scientific Data*, 9(1). doi:<https://doi.org/10.1038/s41597-022-01300-x>.

Waze Partners Help (2025). *WAZE data feed specifications*, Google. Available at:

<https://support.google.com/waze/partners/answer/13458165?hl=en#zippy=%2C retrieving-waze-traffic-data> (Accessed: 26 May 2025).

Weldeghebrael, E. H. (2021). Addis Ababa - ACRC. [online] Available at: <https://www.african-cities.org/addis-ababa/>.

Woldesemayat, E.M. and Genovese, P.V. (2021). Monitoring Urban Expansion and Urban Green Spaces Change in Addis Ababa: Directional and Zonal Analysis Integrated with Landscape Expansion Index. *Forests*, 12(4), p.389. doi:<https://doi.org/10.3390/f12040389>.

World Bank. (2019). Transforming Addis Ababa : Integrated Strategic Development : First Phase - Multi-Sector Diagnostic. [online] Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/339201623400628489/first-phase-multi-sector-diagnostic> [Accessed 8 Jun. 2025].

Zheng, Z., Yang, Z., Wu, Z., & Marinello, F. (2019). Spatial Variation of NO₂ and Its Impact Factors in China: An Application of Sentinel-5P Products. *Remote Sensing*, 11(16), p.1939. doi:<https://doi.org/10.3390/rs11161939>.